



Basic pneumatics (II)



Universidad de Oviedo
Área de Mecánica de Fluidos

Escuela Politécnica de Ingeniería (EPI) de Gijón
4º curso – Grado en Ingeniería Mecánica

**FLUID POWER: HYDRAULIC
AND PNEUMATIC TECHNOLOGIES**



Chapter 3

<https://www.innova.uniovi.es/innova/campusvirtual/campusvirtual.php>



Outline.

- 3.1. Compressed air and pneumatics.
- 3.2. Basic pneumatic system.
- 3.3. Compressors.
- 3.4. Air treatment.
- 3.5. Air distribution.
- 3.6. Actuators. Cylinders
- 3.7. Pneumatic valves.
- 3.8. Vacuum.
- 3.9. Bibliography and additional material.



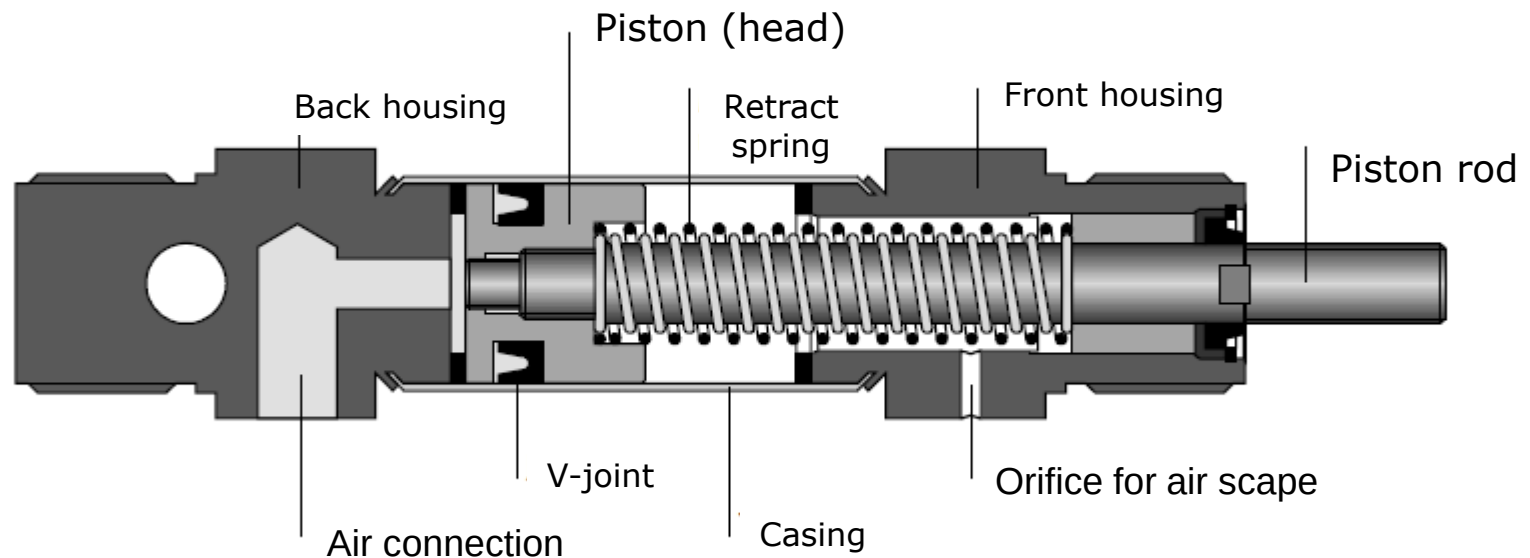
3.6. Actuators. Cylinders.

- ❑ The pneumatic cylinder is very often the final element in the system and delivers the useful power as an output. Its objective is to generate the needed work during its linear movement, extent or retract strokes.
- ❑ The useful work is produced by transforming the pneumatic energy (pressure) into mechanical one (force with a given speed).
- ❑ Types:
 - Simple stroke cylinder.
 - Double stroke cylinder.
 - Special cylinders:
 - With blockage unit and anti-spin system.
 - Parallel and double rod.
 - Double-acting with through piston rod and rod-less cylinders.
 - Plane cylinders and multi-position ones.
 - Sliding table units.
 - Compact and adjustable cushioning cylinders.
 - Hydro-Pneumatic units.
 - Rotating cylinders.



3.6. Actuators. Cylinders.

- ❑ In the **simple stroke cylinders**, the piston head receives the pressurized air in only one side. These cylinders can only transfer energy in one stroke (working stroke).
- ❑ The retract stroke is performed either by the spring inside the cylinder or by an external force (gravity, for instance).
- ❑ There are three main types: piston cylinders (the most common ones), membrane cylinders and bellows cylinders.





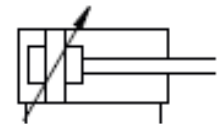
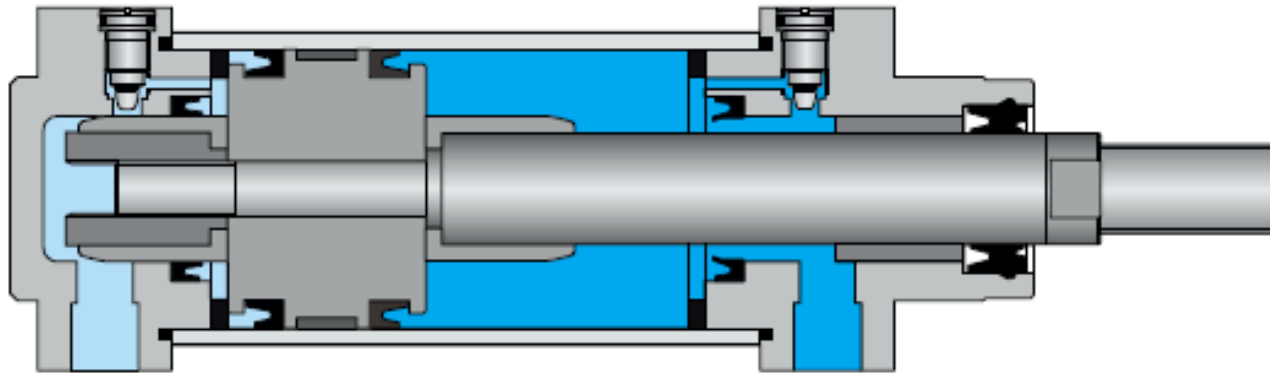
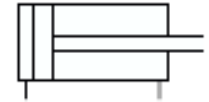
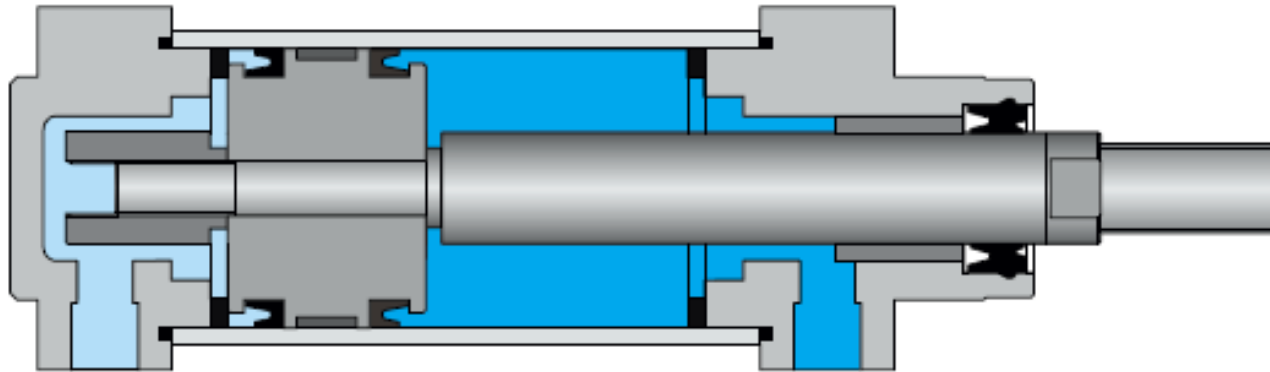
3.6. Actuators. Cylinders.

- ❑ In the **double stroke cylinders**, the piston receives the pressurized air in both sides. The cylinder produces an effective work in both movements (extent and retract stroke).
- ❑ In the simple rod cylinders, the extent force is bigger than the force in the retract movement (bigger surface in one side and in the other, due to the rod).
- ❑ They are built with cushioning when there are high weights moved or forces to be applied, to avoid the shock at the end of the stroke. Such cushioning is often done with an air restriction, that can be externally regulated and that controls the air flow, reducing it.
- ❑ Among the double stroke cylinders, four basic types are found: simple-rod piston cylinders, double rod cylinders, trough rod cylinders, multi-positional cylinders, impact cylinders. Again, the mostly used one is the piston cylinder with simple rod.



3.6. Actuators. Cylinders.

Double stroke cylinder.

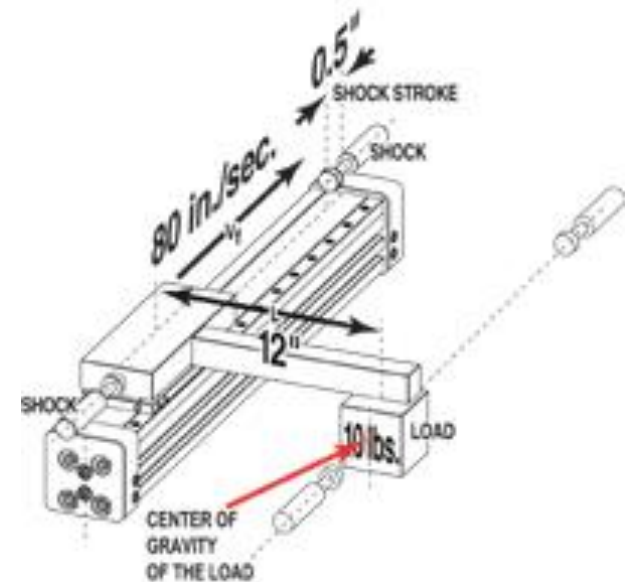


Double stroke cylinder with variable end of stroke cushion.



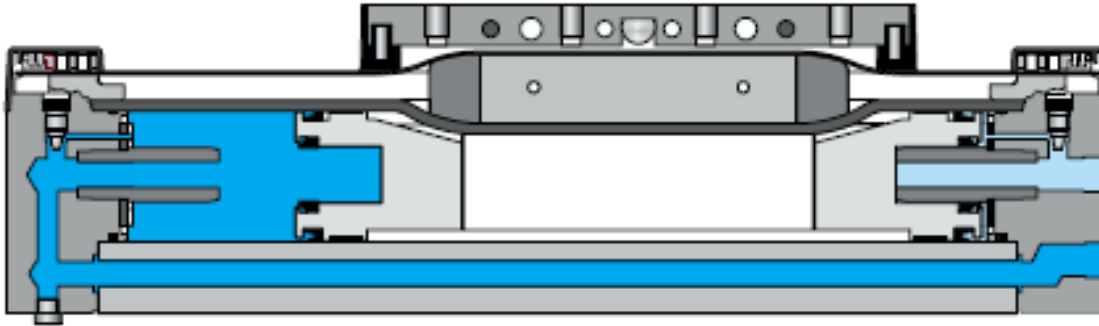
3.6. Actuators. Cylinders.

- ❑ In the **rod-less cylinders**, pressurized air is applied alternatively in both sides of a mobile element, as in a double stroke cylinder. Their use is adapted for small working places or where the extent rod promotes some space incompatibilities.
- ❑ They are shorter than the standard ones (double stroke cylinders) and work within a rail or guide, so there is no possible twist of the rod. Besides, the movement is done in the whole length of the stroke, that can be increased up to 10 m.
- ❑ Among the simple stroke cylinders, three different options are available: airtight band cylinder with ribbed sleeve, magnetic coupled cylinders and belt (or cable) cylinders.

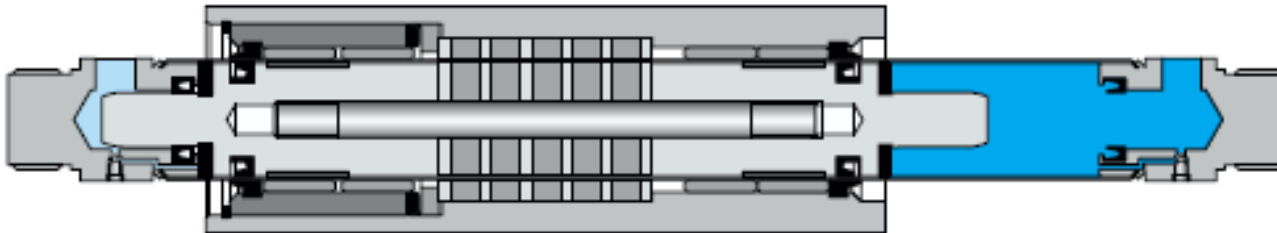




3.6. Actuators. Cylinders.



**Airtight band
cylinder**



**Magnetic coupling
cylinder**



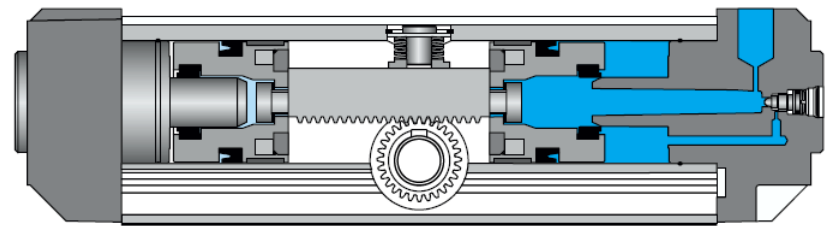
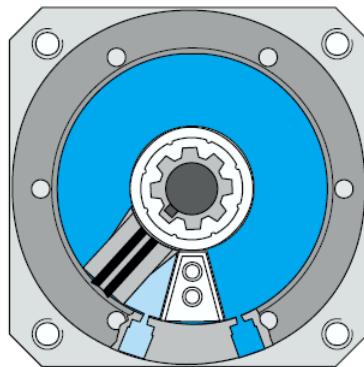
**Cable or belt
cylinder**



3.6. Actuators. Cylinders.

- ❑ In the **rotating cylinder**, the rod has a teeth structure that is in gear with a pinion wheel. The rod movement (linear) is transformed into a rotating one. Rotating angles from 0 to 360° are possible. These kind of cylinders bear torques between 0.5 and 150 Nm at 6 bar, depending on the piston diameter.
- ❑ In the **oscillating cylinder**, the pressurized air moves a sliding vane and this movement is transmitted to the shaft. Rotating angles from 0 up to 270° become possible. These cylinders bear torques between 0.5 and 20 Nm at 6 bar, depending on the size of the vane.

Oscillating
(sliding vane)
cylinder

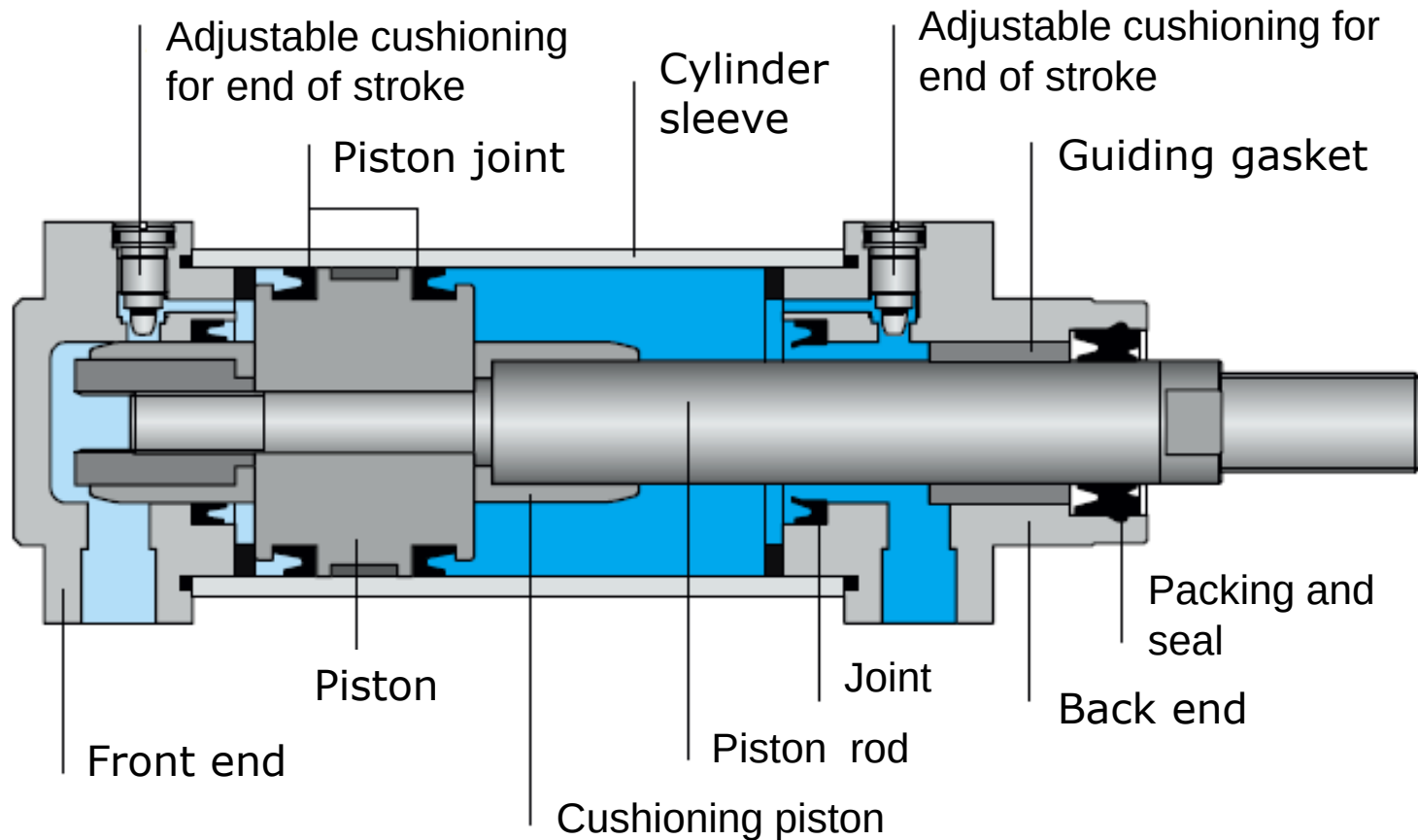


Rotating cylinder



3.6. Actuators. Cylinders.

Cylinders internal constitution.

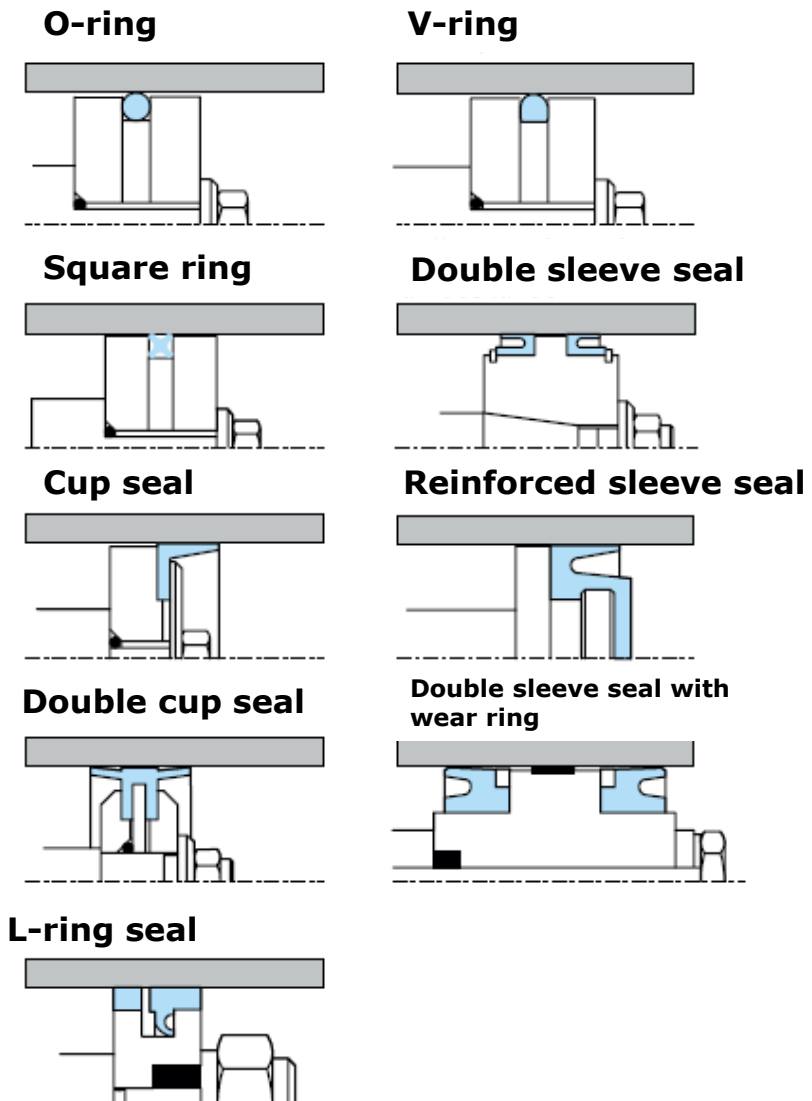


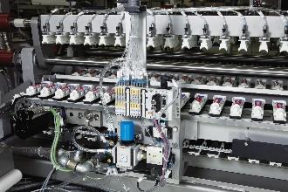


3.6. Actuators. Cylinders.

Types of joints:

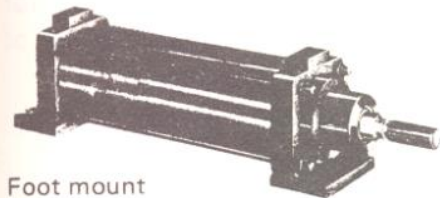
- ❑ The pneumatic cylinders need some **airtightness** in the contact of mobile and static parts. Such depends on the joints or auxiliary rings mounted between pistons or rods and the casing.
- ❑ The criteria to be taken into account for each mounting are: dimensions, adjustment, wear resistance, useful life, resistance to chemical agents, etc....
- ❑ Widespread materials for the joints are: Perbunan (nitrile rubber), viton (Fluor elastomer), teflon.



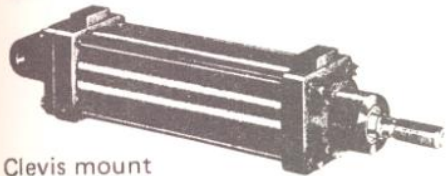


3.6. Actuators. Cylinders.

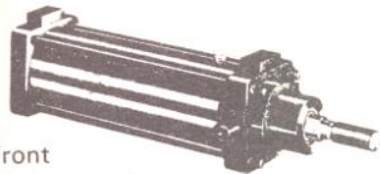
Cylinder mounts (basic).



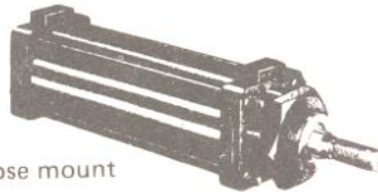
Foot mount



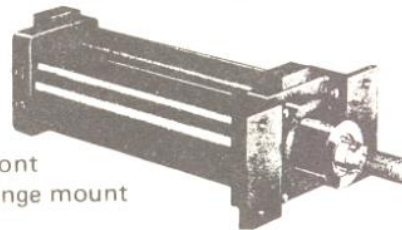
Clevis mount



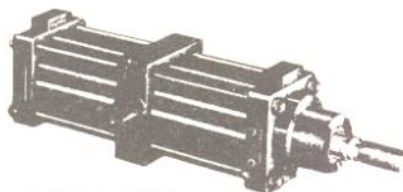
Front pivot mount



Nose mount



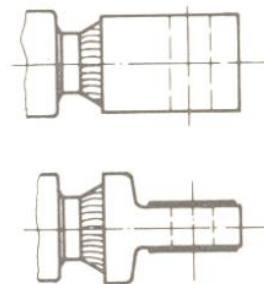
Front flange mount



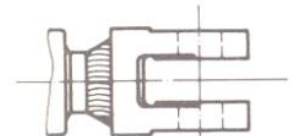
Trunnion mount

Different practical configurations. In the next slide, a wider range of options are shown.

Simple lugs depending of cyl. dia.



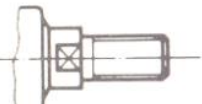
Fork heads

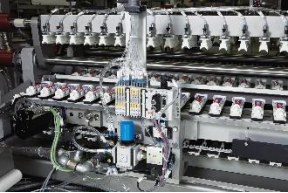


Bearing eye heads



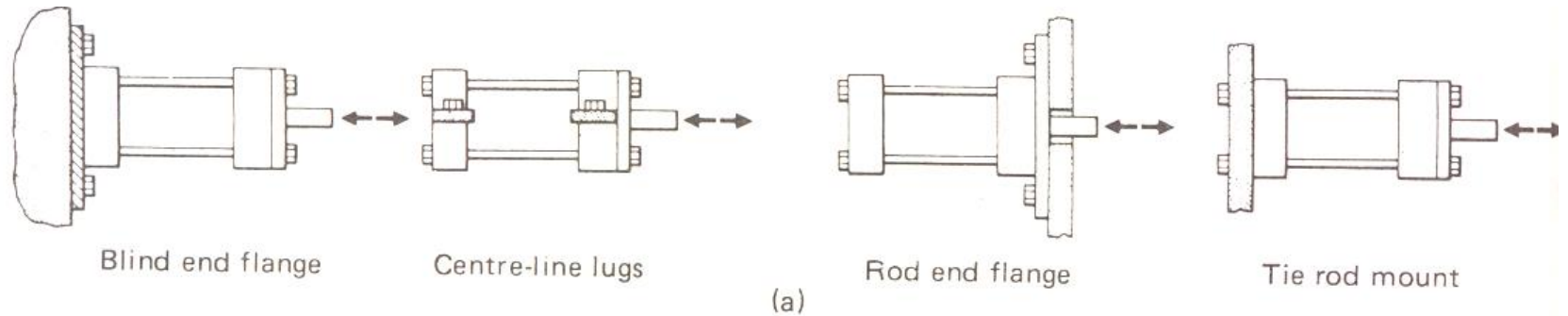
Thread on rod end



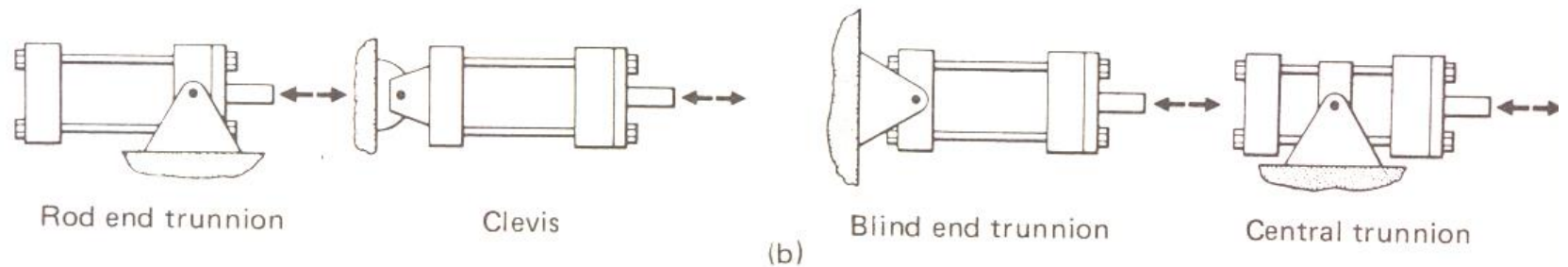


3.6. Actuators. Cylinders.

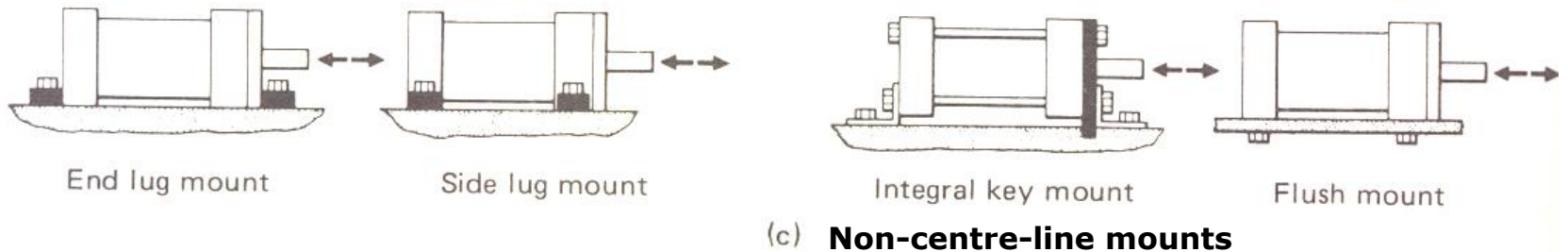
Cylinder mounts.

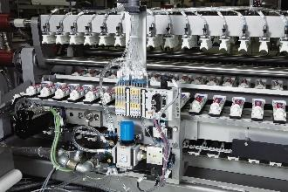


Fixed centre line



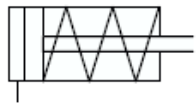
Pivoted centre line



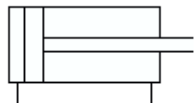


3.6. Actuators. Cylinders.

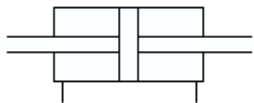
Linear elements:



Single stroke cylinder.



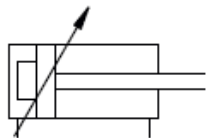
Double stroke cylinder (simple).



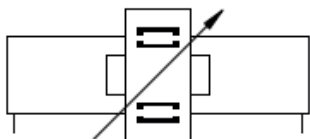
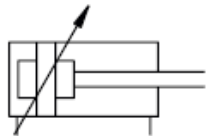
Through piston rod double stroke cylinder.



Double stroke cylinder with fixed cushioning.

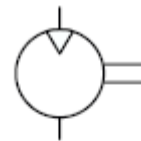


Double stroke cylinders with variable cushioning (simple and double cushioning).

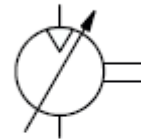


Rod-less cylinder (magnetic coupling).

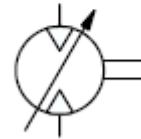
Rotating elements:



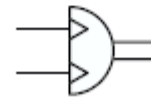
Pneumatic motor with constant cubic capacity and fixed flow direction.



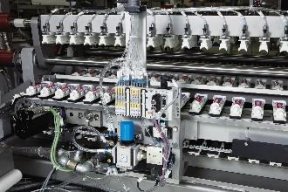
Pneumatic motor with variable cubic capacity and fixed flow direction.



Pneumatic motor with variable cubic capacity and two possible flow directions.



Pneumatic oscillating motor (cylinder).



3.6. Actuators. Cylinders.

Linear actuator selection.

- ❑ **Step 1:** Dimensions of the cylinder head or piston using the **force value**.
 - Determine the **load factor** as a function of the requirements. The loading factor is a security parameter that considers the cylinder acceleration (average speeds: 0.7, high speeds: 0.4-0.5).

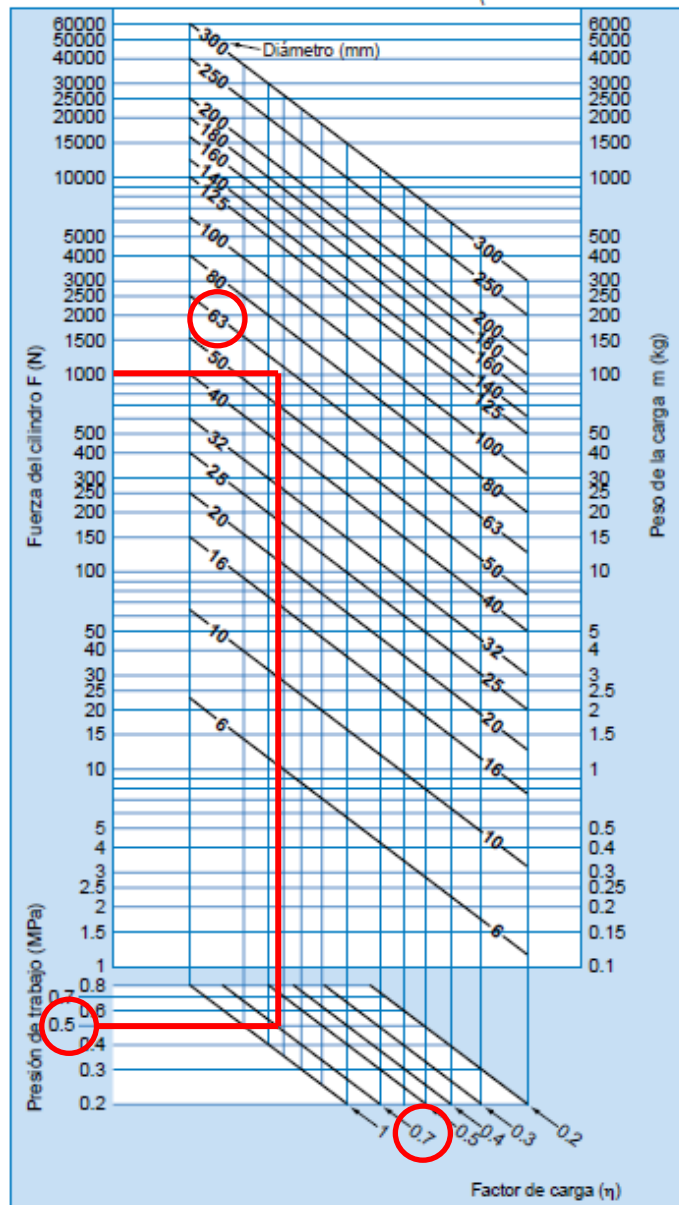
Required working conditions		Load factor
Static conditions (tie up, screwing at low speed, etc.)		≤ 0.7 (70% or lower)
Dynamic conditions	Horizontal movement of the load	≤ 1.0 (100% or lower)
	Vertical movement of the load	≤ 0.5 (70% or lower)

- Fix the working pressure. As a general rule, at 85% of the delivery pressure. Meaning this, **a load efficiency: $\eta = 0.85$**
- Determine the direction of the force in the cylinder movement (extent or retract movement).

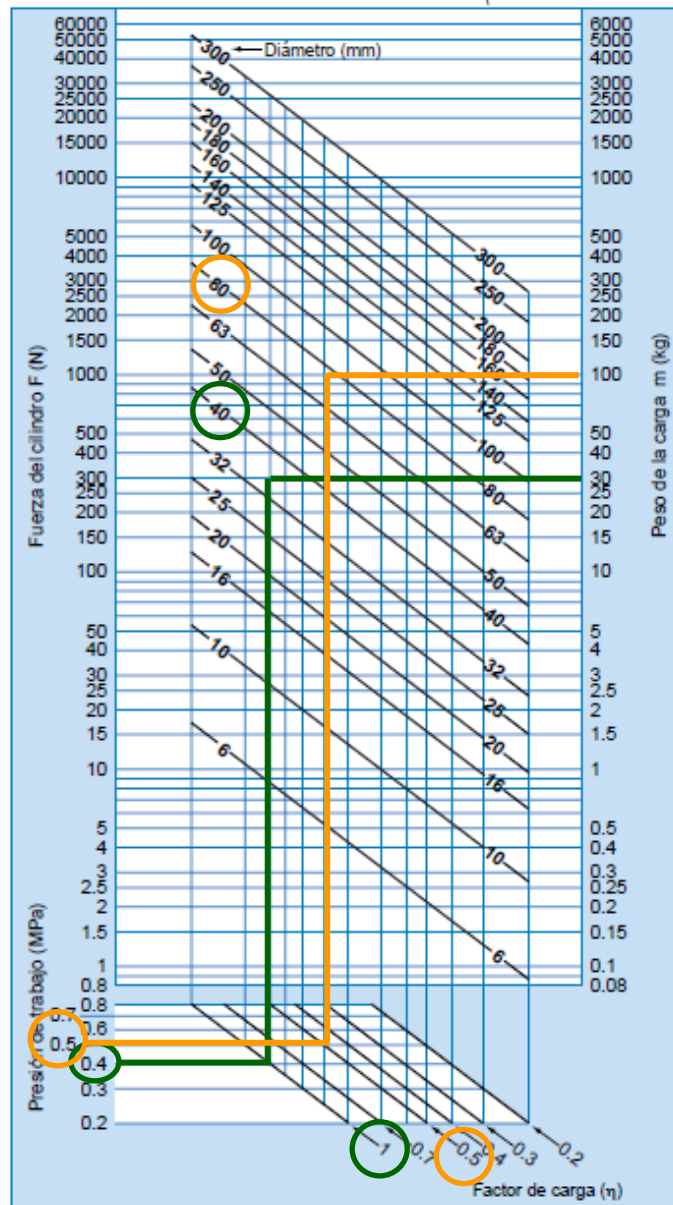


3.6. Actuators. Cylinders.

<Gráfico 1> Fuerza del cilindro en el lado de extensión (Cilindro de doble efecto)



<Gráfico 2> Fuerza del cilindro en el lado de retracción (Cilindro de doble efecto)



Graph 1.
Applied force for the
extending.

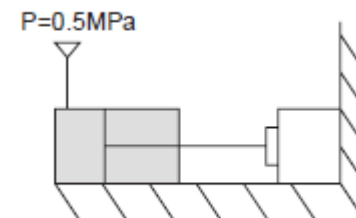


Fig. 1

Graph 2.
Applied force for the
retracting.

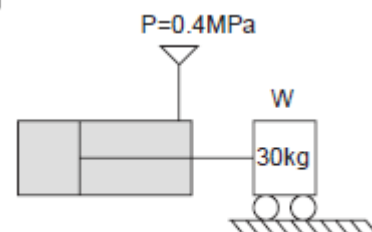
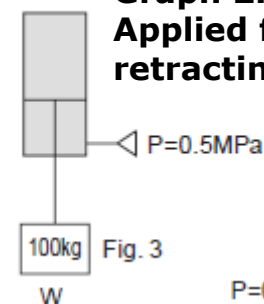
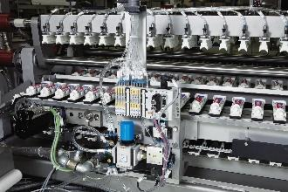


Fig. 2



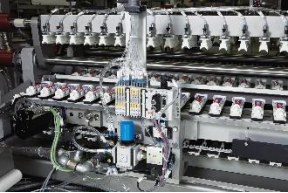
3.6. Actuators. Cylinders.

- ❑ **Example 1:** If a minimum force of 1000 N would be needed to keep the working piece as in Fig. 1, as it corresponds to the extending movement, use Graph 1 with a load factor of 0.7 and a working pressure of 0.5 MPa to obtain the piston diameter. **Solution:** 63 mm.
- ❑ **Example 2:** To move a load with a weight of 30 kg horizontally as in Fig. 2, as the load is the same for the extend and retract movement, using Graph 2, that is equivalent to the retract movement with a lower force. For a load factor of 1 and a working pressure of 0.4 MPa, obtain the diameter. **Solution:** 40 mm.
- ❑ **Example 3:** To move a load of 100 kg vertically and upwards as in Fig. 3, use Graph 2 to determine a load factor of 0.5 and a working pressure of 0.5 MPa and obtain the diameter. **Solution:** 80 mm.



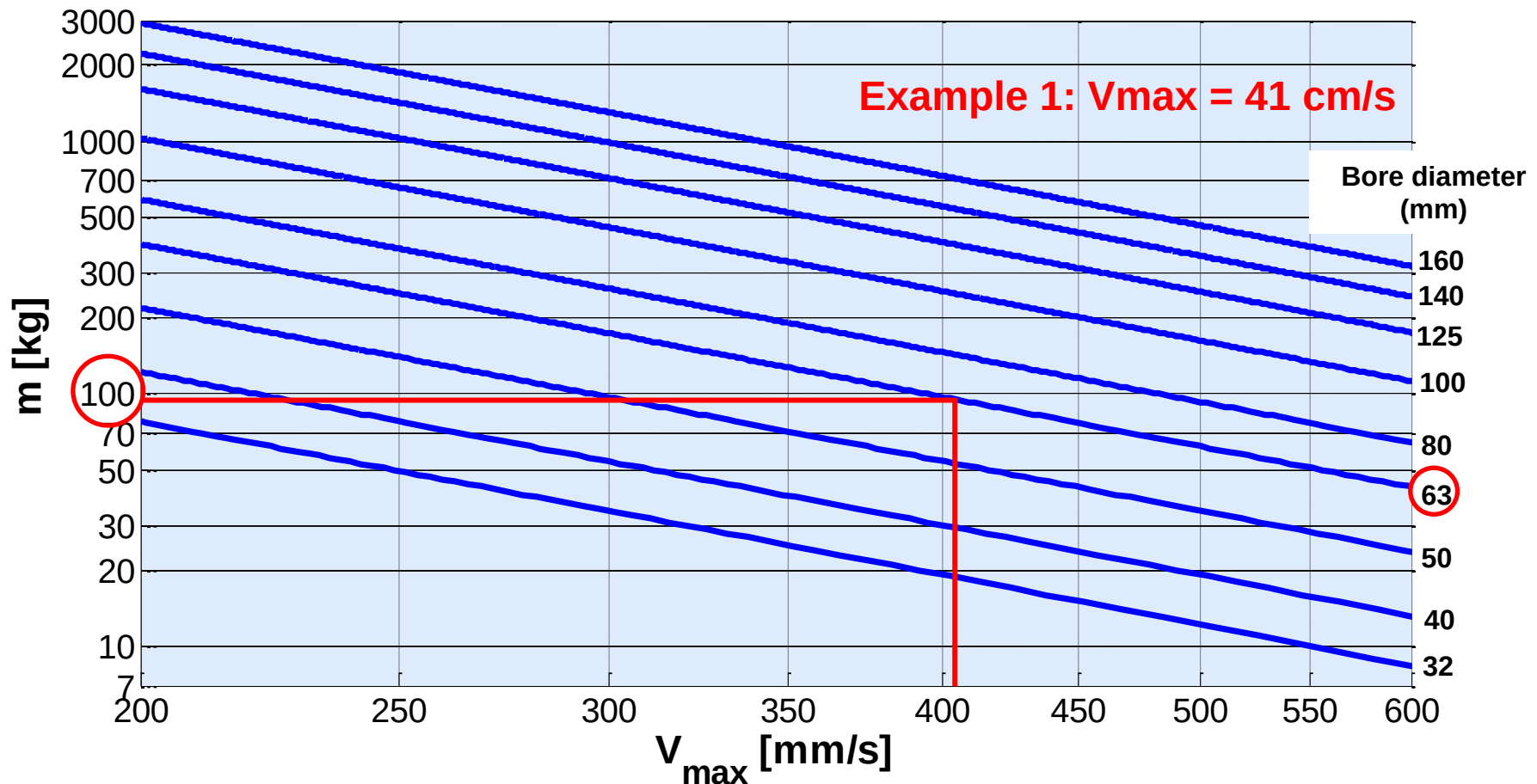
3.6. Actuators. Cylinders.

- ❑ **Step 2:** Consider the **impacts** at the end of the stroke:
 - When there is an external end of stroke (hydraulic end of stroke, etc.) to absorb the impact, it must be selected a cushioning system with enough absorption capacity.
 - Without any specific system, check in the following graph the cylinder's self capacity.
 - **Elastic cushioning:** The plastic "uretan" it is used to prevent the metallic contact between the cylinder head and the casing (cylinder's housing).
 - **Pneumatic cushioning:** The exiting air it is compressed just before the end of stroke and the generated reaction force absorbs the kinetic energy of the load, allowing a smooth load stoppage.



3.6. Actuators. Cylinders.

- **Graph:** Maximum speed check of a cylinder that has to move a given load with mass “m” as a function of the diameter of the bore, so that the cushioning would be effective (no additional cushioning required).





3.6. Actuators. Cylinders.

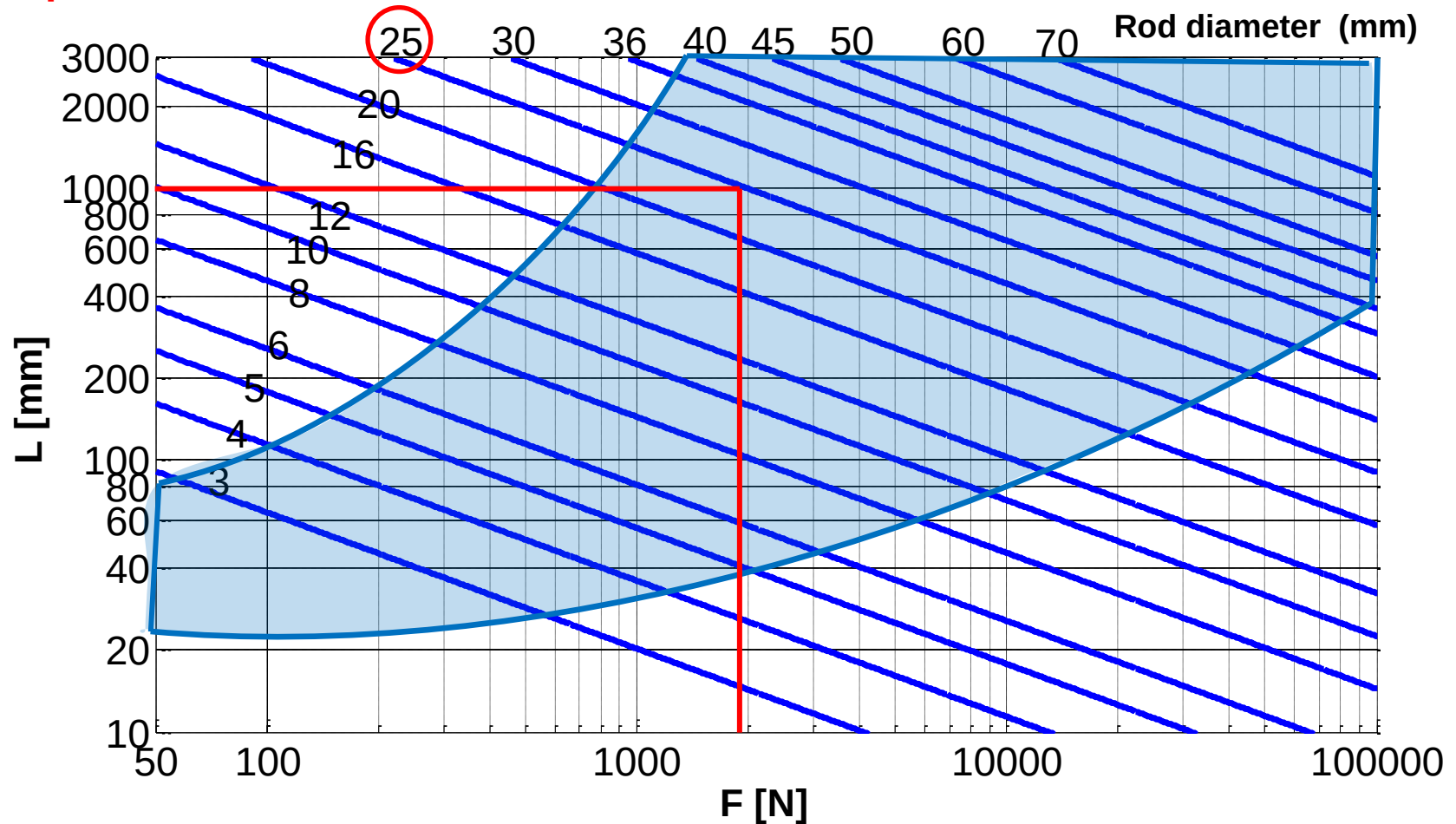
- ❑ **Step 3:** Bear in mind the structural limits, such as the **buckling limit** and the maximum bearable **radial forces**.
 - Buckling is defined as “To bend, warp, or crumple, as **under pressure** or heat”. Buckling is characterized by a sudden sideways failure of a structural member subjected to high compressive stress, where the compressive stress at the point of failure is less than the ultimate compressive stress that the material is capable of withstanding. In a cylinder bore, the buckling depends on the value of the load, the bore dimensions, its material, length and kind of mount.
 - In practise, there are unexpected misalignments, load lack of symmetry, that preclude a given torque on the cylinder joints, seals and gasket. These may produce a quick wear and shorten the useful life of the cylinder. The check against radial unbalanced loads depends on the cylinder's length, the kind of mount, the load on the gasket, the total cantilever distance and the maximum load at the bore end.

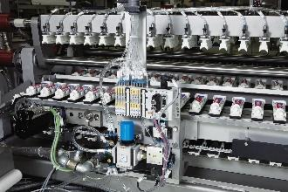


3.6. Actuators. Cylinders.

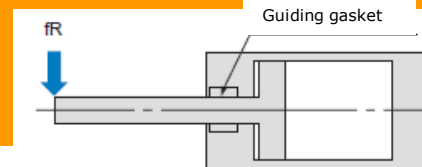
- **Bucking graph:** Maximum stroke (L) for a given rod diameter and required force (N).

Example 1: rod of 25 mm and stroke of 1 m





3.6. Actuators. Cylinders.

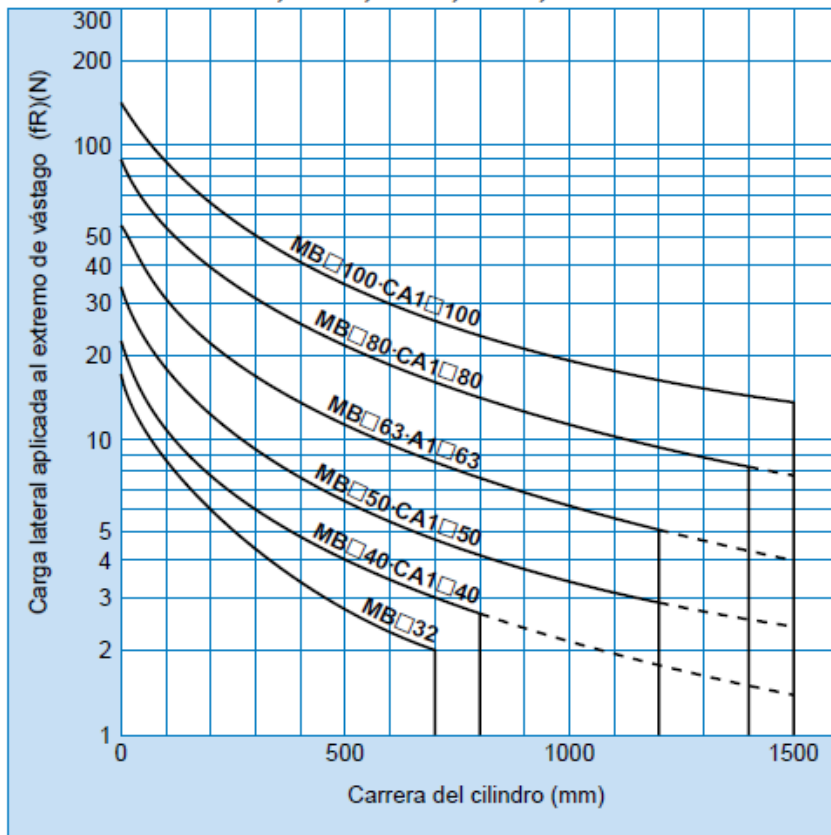


- Graph with the axial loads: Maximum loads as a function of the stroke and diameter of the cylinder.

<Gráf. 10>

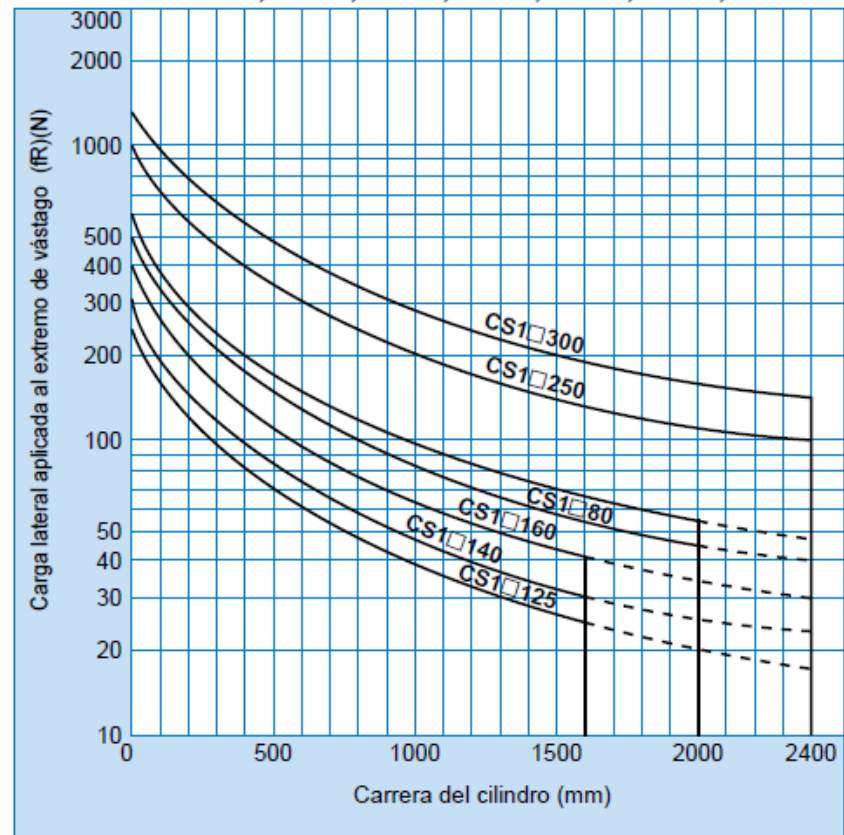
Series MB/ø32, ø40, ø50, ø63, ø80, ø100

Serie CA1/ø40, ø50, ø63, ø80, ø100



<Gráf. 11>

Serie CS1/ø125, ø140, ø160, ø180, ø200, ø250, ø300





3.6. Actuators. Cylinders.

- ❑ **Step 4:** Calculation of the **air consumption** at the cylinder and the required **air volume**:
 - In the air systems, the volume of air needed is due to the movement of the cylinders, together with the cylinders' connections and the opening valve, whenever this valve is operated. This total volume is needed to obtain the compressor dimensions and obtain an estimation on the costs.
 - The air consumption in a cylinder is obtained as: piston surface by length of the stroke by number of strokes by the absolute pressure (Boyle-Mariotte law is applied to know the Normal conditions air volume).

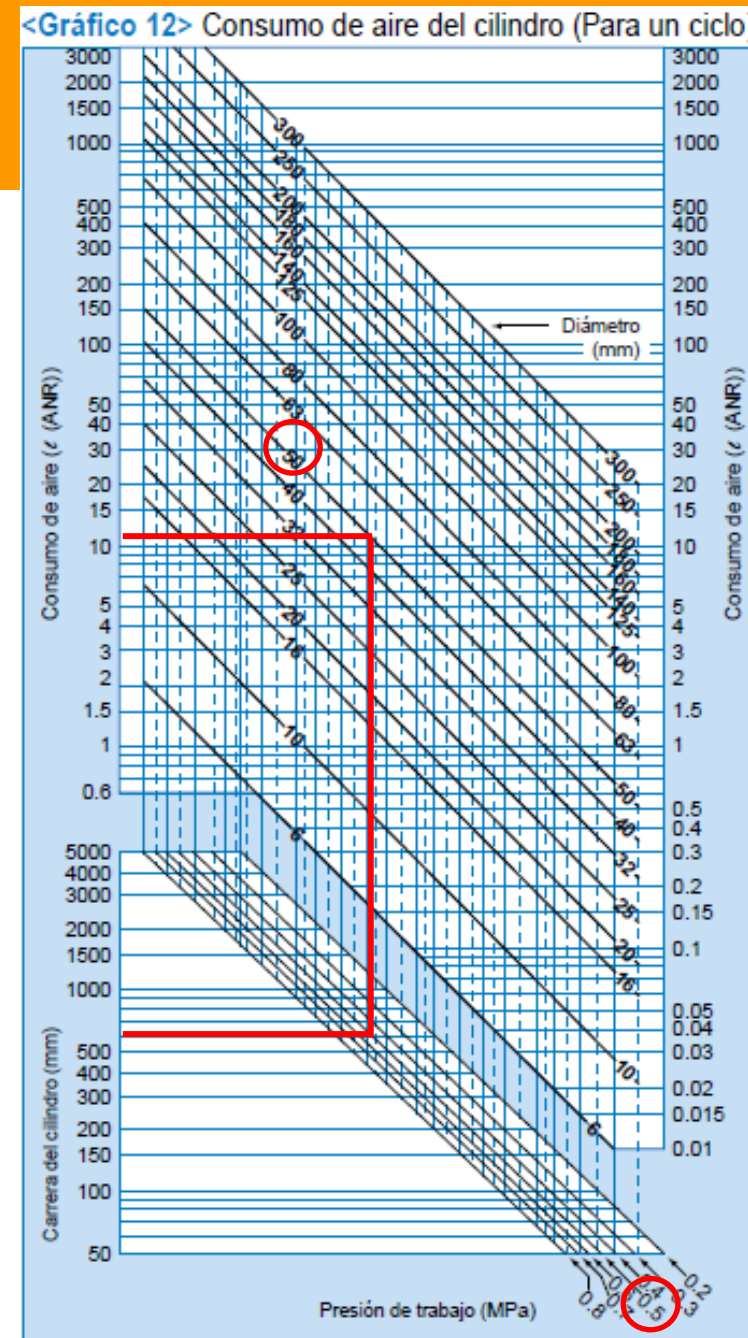
$$V [NL] = \pi \frac{(D_E [mm])^2}{4} \cdot L [mm] \cdot \frac{P [MPa] + 0.1013}{0.1013} \cdot 10^{-6}$$

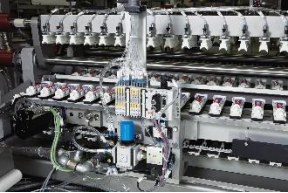
- Also the leakage losses on the pipes must be estimated.



3.6. Actuators. Cylinders.

- ❏ **Example 4:** Consider the use of 10 air cylinders with a diameter of 50 mm and a stroke of 600 mm for a pressure of 0.5 MPa, ¿what is the air requirement at 5 cycles per minute?
- The piston surface is 1963 mm². The air volume is, then 1963 600=1177800 mm³. The amount of atmospheric (normal conditions) air is, then: 1.178 by (0.6013/0.1013) = 7 NI. The whole cycle comprises the extend and retract movement. Therefore, the air consumption is 14 l. (in the graph, it is read around 13 l, that is, for the whole cycle).
 - Then, multiplying by 5 cycles, the value of 70 NI is found for a given cylinder (14x5).
 - As, there are a total of 10 cylinders, the final consumption is 700 NI.

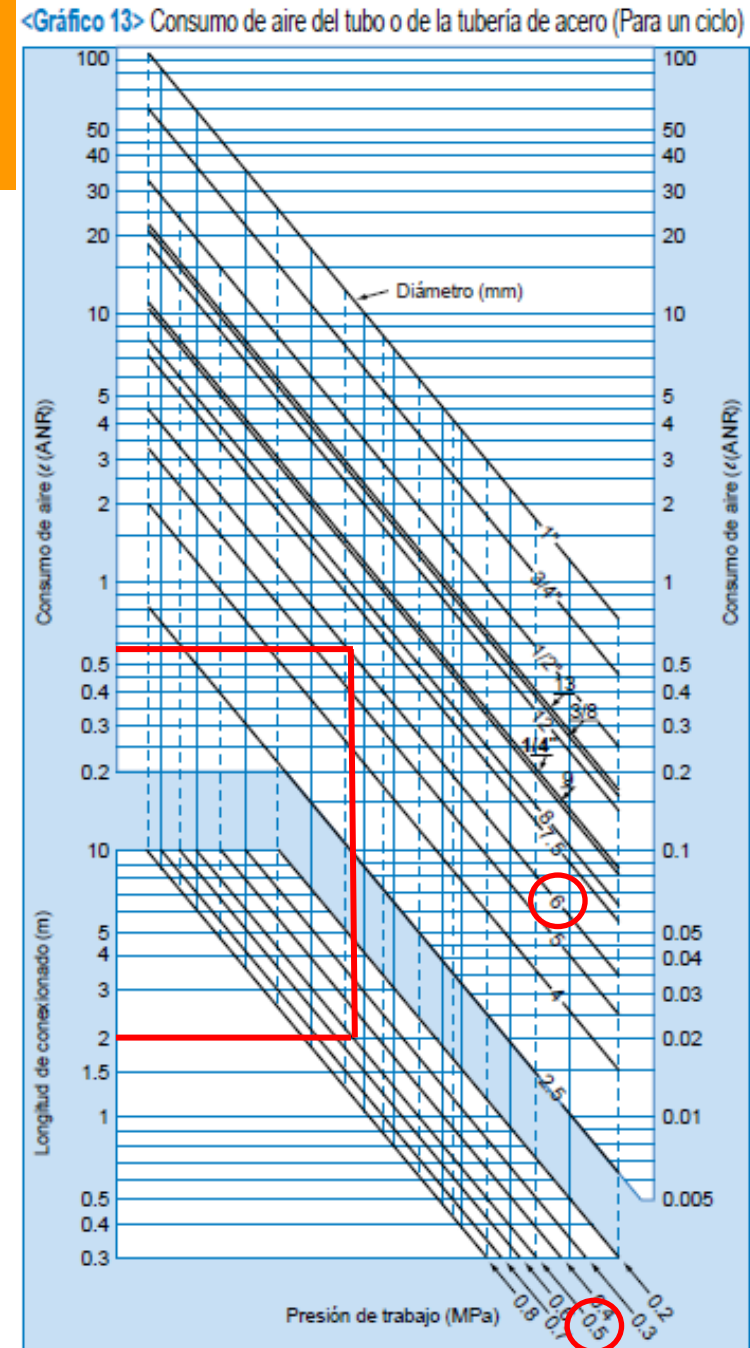




3.6. Actuators. Cylinders.

❏ **Example 5:** Considering the previous problem, with a pipe of 2 m and 6 mm diameter for the connection between cylinders and the directional valve, what additional consumption would be calculated for a good working conditions?

- With the graph for 0.5 MPa and connection length of 6m, finding the crossing with pipes with a diameter of 6 mm, and additional consumption of 0.56 NI is found.
- Therefore, multiplying by 10 cylinders and 5 cycles, the total is: 28 NI, to be summed up to the previously obtained 700.
- In this case, the final value corresponds to a value of 4% in losses of the useful value.





3.6. Actuators. Cylinders.

- ❑ **Step 4-bis:** Calculation of the **air consumption** at the cylinder and the required **air volume**:
 - Alternatively, it can be estimated the air volume required for a given cylinder when a speed for the cylinder is known. This calculation is needed also to choose the diameter of the set of pipes after the system's main valve or after the FCL system.
 - The air consumption in a cylinder is obtained as: piston surface by maximum speed by the absolute pressure (again, the Boyle-Mariotte law is applied to know the Normal conditions air volume).

$$Q(Nl) = 60 \pi \frac{(D_c (\text{mm}))^2}{4} v (\text{mm/s}) \frac{P (\text{MPa}) + 0.1013}{0.1013} 10^{-6}$$



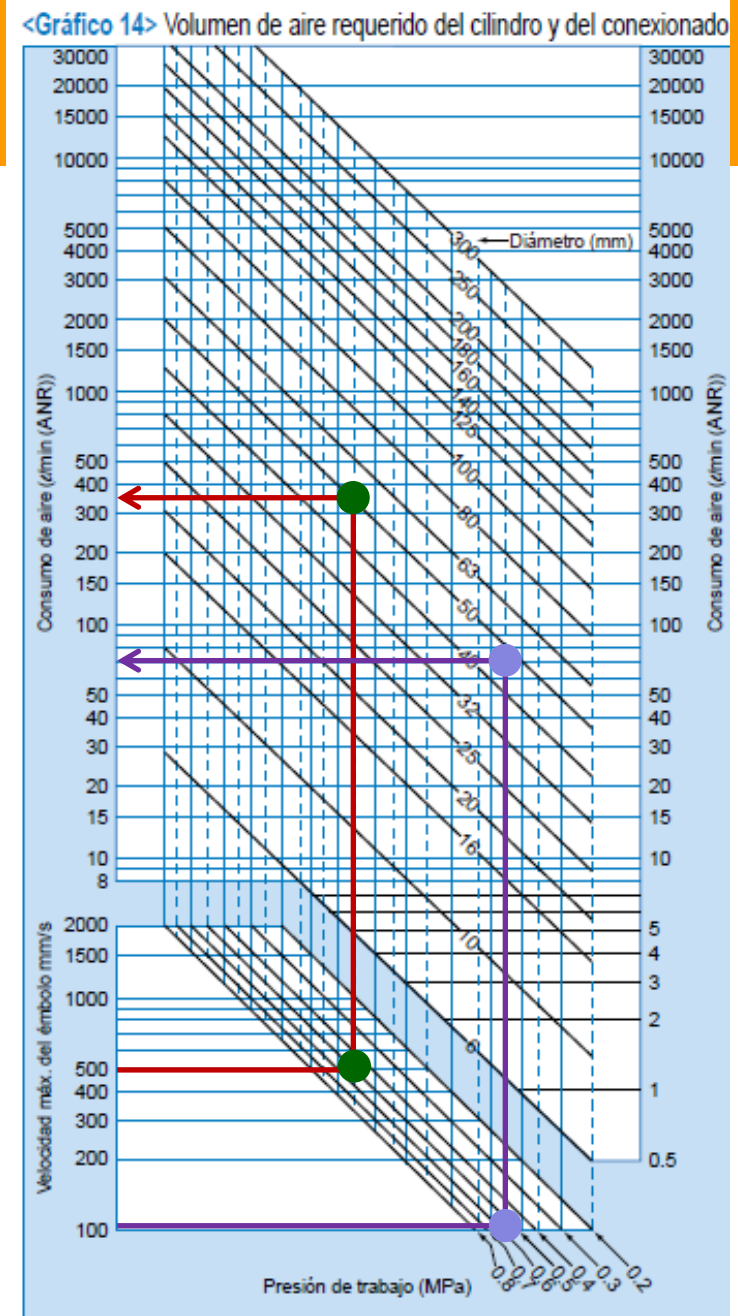
3.6. Actuators. Cylinders.

❑ **Example 6:** What is the volume needed to move a cylinder with a diameter of 50 mm, at a pressure of 0.5 MPa, and a speed of 500 mm/s?

- Using the graph for 0.5 MPa and maximum piston speed of 500 mm/s, and finding the intersection with the diameter of 50 mm, a consumption of 350 NI is obtained.

❑ **Example 7:** Use the same procedure as for example 6 with the data of example 4.

- In the previous case, with 5 cycles/min, it means that each cycle is done in 12 s. Considering 6 s for the extent, the speed would be $600 \text{ mm}/6 \text{ s} = 100 \text{ mm/s}$. For the graph with 100 mm/s, it is obtained (approximately) the previous value of 70 NI.





3.7. Pneumatic valves.

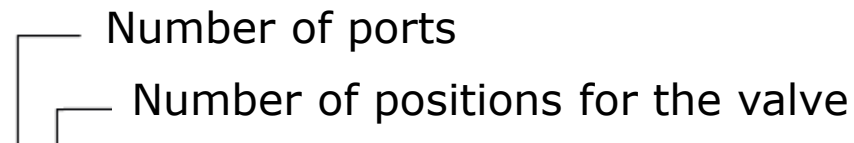
- ❑ The elements used to control the flow in pneumatic circuits are the **valves**. It may be defined a valve as the element in charge of regulate or control the starting up, shut-off and direction together with the pressure, driven by a compressor, a vacuum pump or an accumulator.
- ❑ Apart from the type of construction arrangement (ball, poppet, plug, etc.) the main goal of them is the **function** that can be obtained from them. In such way, according to the different functions, a global classification could be:
 - **Directional-control valves:** determine the air flow path.
 - **Shut-off:** cut the air pass to all or a set of directions.
 - **Pressure-control valves:** used to control the pressure for a safety and efficient operation of the different components in pneumatic circuits.
 - **Flow-control valves:** Theses valves are used to regulate the air flow rate in a given pipe.



3.7. Pneumatic valves.

- ❑ The direction-control valves are **control elements, that depending on the inlet value produce a defined outlet.**

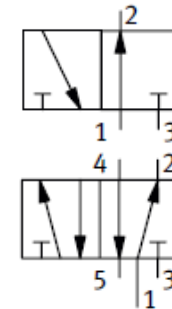
- ❑ **Naming the ports and positions:**



Valve 3/2, normally opened

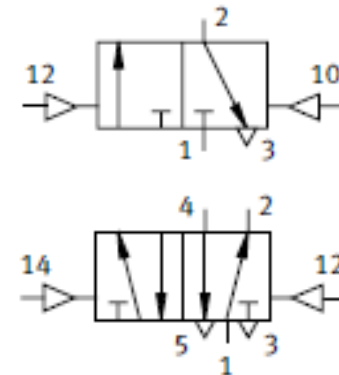
Valve 5/2

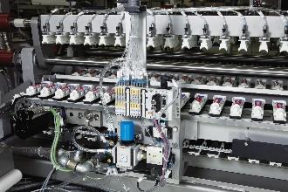
Flow rate from 1 → 2 and 4 → 5



- ❑ **Ports naming:**

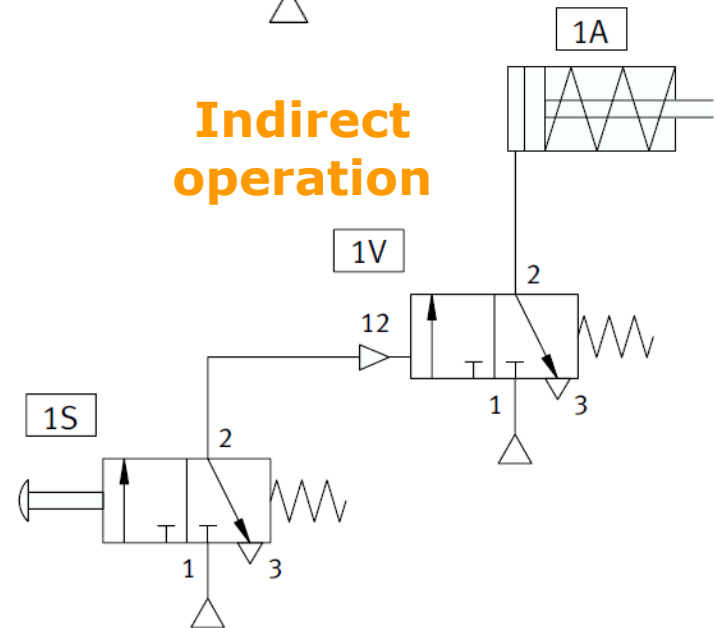
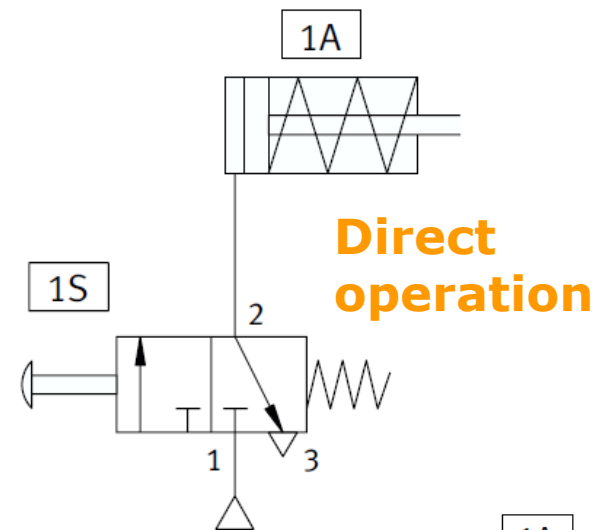
- **Working ports:** 1 (pressurized air); 2,4 (working ports); 3,5 (air exit to atmosphere).
- **Operation points:** 10 (if there is signal here, 1 is closed and 2 is connected to exit), 12 (air passes from 1-2), 14 (air passes from 1-4).

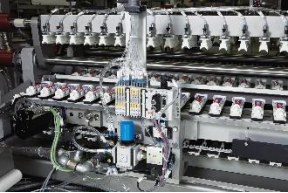




3.7. Pneumatic valves.

- ❑ To change from one to other position in the direction-control valves, a given signal is needed. The operation can be **direct or indirect**.
- The valves with direct operation provides an operation directly placed on the valves (at any side of the valve). The operation may be manual or mechanical. The usual manual operations are: push button, knob, pedal, etc. The mechanical operations are: plunger, rollers, etc.
- The indirect operation is done using either pneumatic or electric signal. The pneumatic pilot may also be done by a pressure impulse (positive operation) or by a pressure decrease (negative operation). The electrical operation is performed using electro-magnets.





3.7. Pneumatic valves.

- ❑ The **operation type** is chosen according to the system requirements:

- Manual operation.
- Mechanical operation.
- Compressed air operation.
- Electric operation.
- Mixed operation.

- ❑ Besides, the **final goal** of the operation signal can be:

- Move the valve.
- Reposition it.
- Centre the valve.

Hand operated

General manual



Lever



Push button



Pedal



Pull button



Treadle



Push/pull button



Rotary knob



Mechanical operated

Plunger



Spring normally as a return

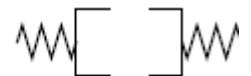


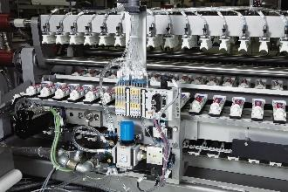
Roller



Spring centred (both sides)

Uni-direction or one way trip





3.7. Pneumatic valves.

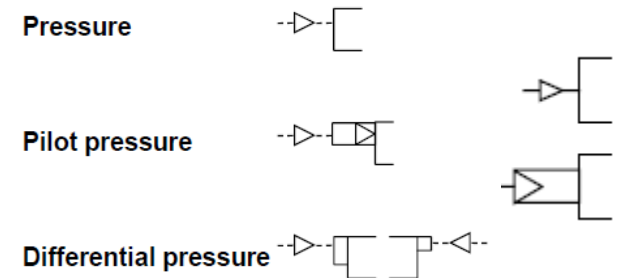
❑ In addition of the type of valve's acting or operation, also the **time to move** can be considered:

- Permanent acting, stable signal. The valve can be manually, mechanically, electrically or with pneumatic operation along a given period of time. The change to neutral position is usually done by a spring (**Mono-stable**).
- Impulse or sudden operation. The valve is moved by a short time signal (impulse) and it is kept in that position until an opposite signal is received (**Bi-stable**).

Basic valve construction

- Poppet valve.
- Plate valve.
- Spool valve.

Pressurized air operated:



Electromagnetic operated:

Solenoid direct



Solenoid pilot



Electromagnetic valve with auxiliary manual override operation

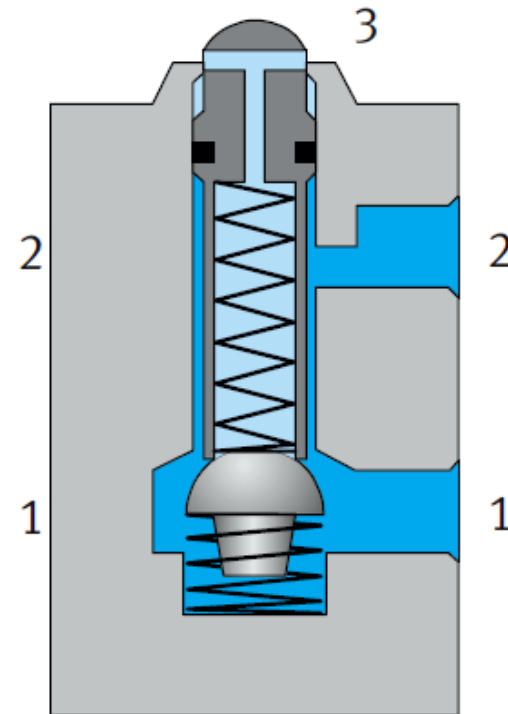
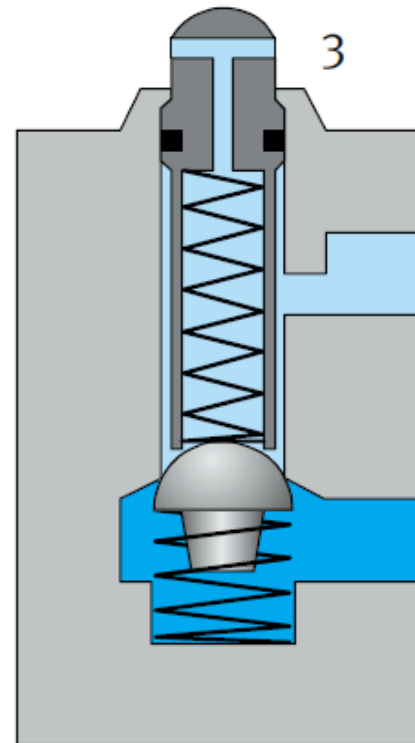
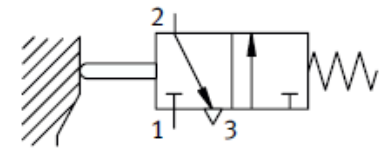
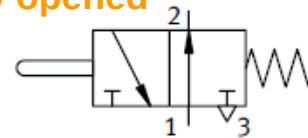




3.7. Pneumatic valves.

- ❑ The **3/2 valves** are often used for the control of simple stroke cylinders and to promote a pneumatic pilot signal. Due to their three ports, they allow two ways for the air flow.
- ❑ They can be normally closed or normally opened, depending on their neutral position.
- ❑ They produce no interferences during commuting.
- ❑ They could have **ball** or **poppet** closing element.
- ❑ Easy and economic design, they are not affected by dust and have a long duty life.

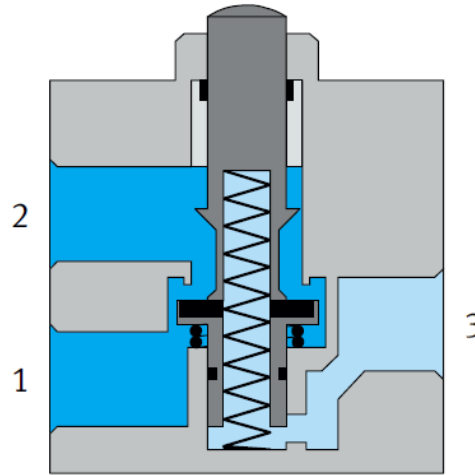
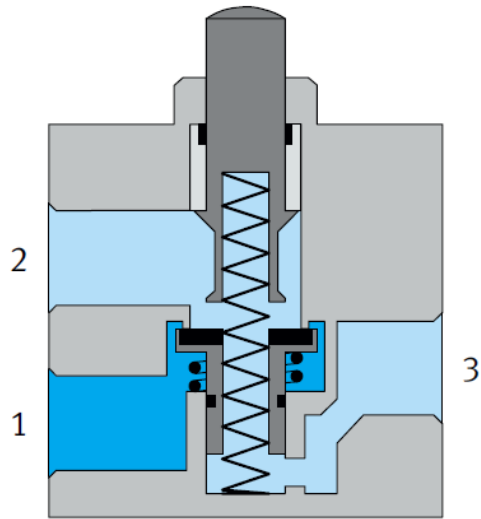
Normally opened



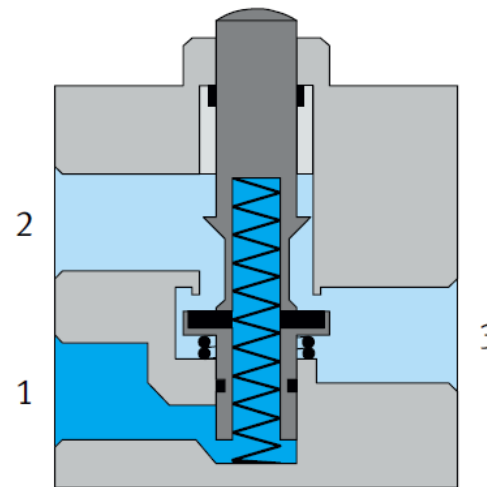
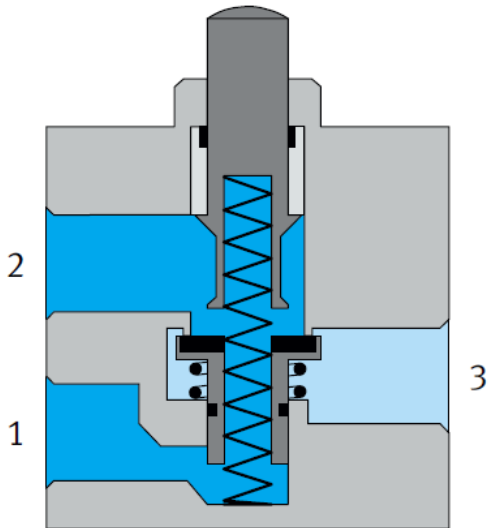
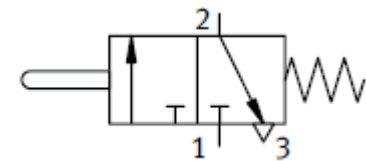
Normally closed



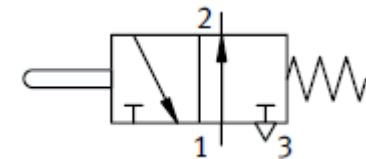
3.7. Pneumatic valves.



**Valve 3/2, plunger ,
normally closed.**



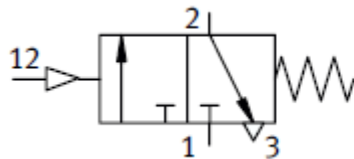
**Valve 3/2, plunger ,
normally opened.**



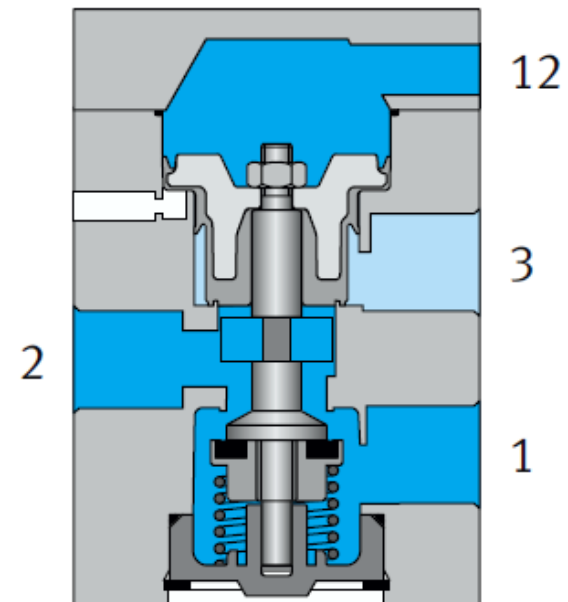
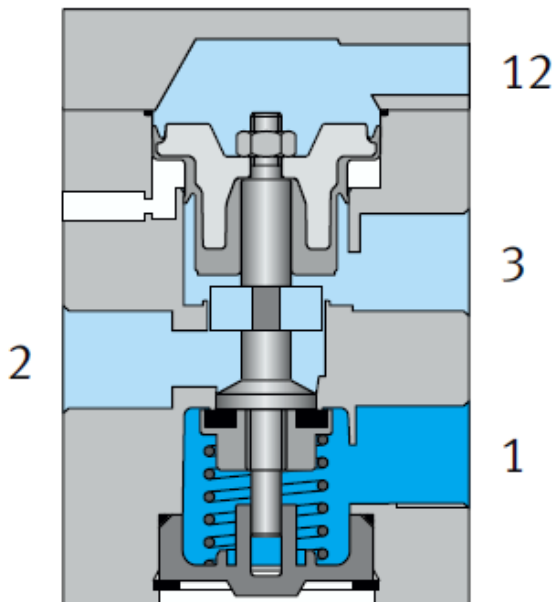
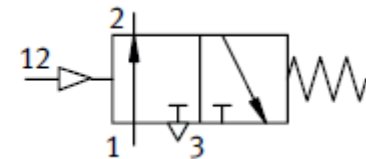


3.7. Pneumatic valves.

- In the case of valves 3/2 **mono-stable** they include a pneumatic operation with pilot connection to allow the valve movement. The internal surface of the piston should be chosen in a way such as for an equal pressure value in the pilot and pressure line, the valve could be moved without any problem.



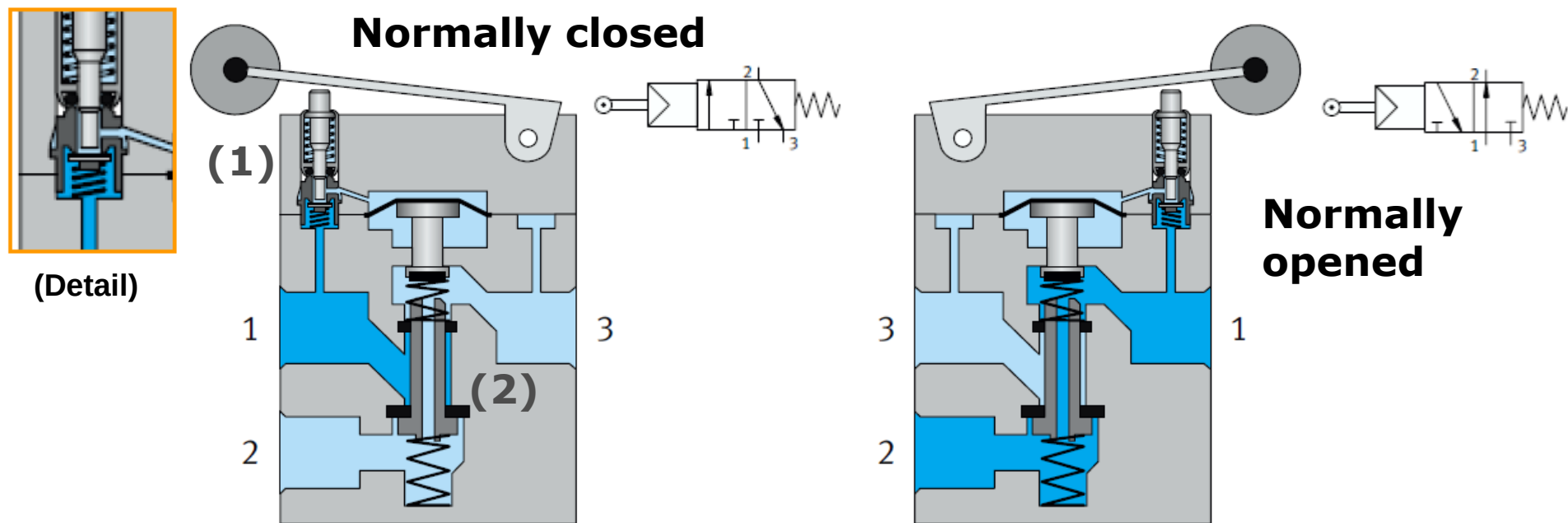
**Valve 3/2,
mono-stable,
normally closed**





3.7. Pneumatic valves.

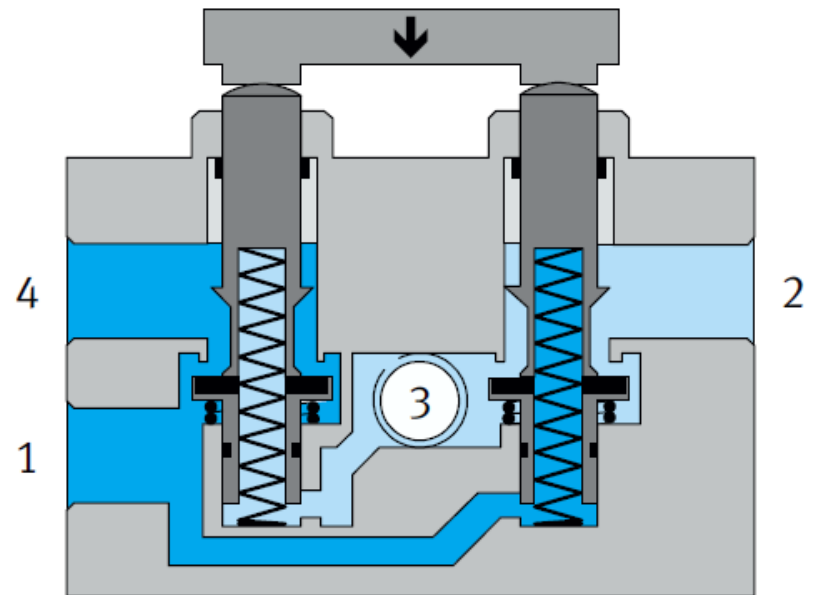
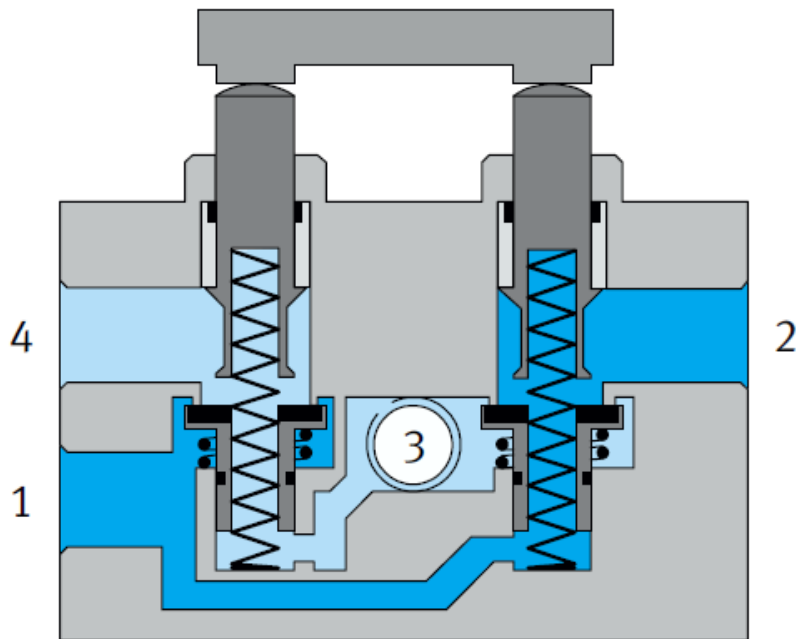
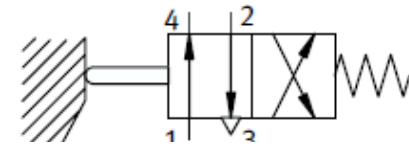
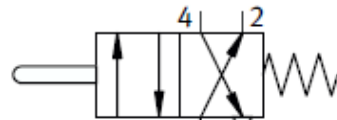
- ❑ Valves 3/2 operated by **roller and servo-pilot**.
- ❑ The servo-pilot allows to strongly reduce the movement forces. It is basically constructed with a two valve composition: (1) the auxiliary servo-piloted valve and (2) the main valve.
- ❑ A small air pipe is activated when the cam hits the rod. The pressure link between the membrane and the valve operation is established and the valve opens.





3.7. Pneumatic valves.

- ❑ The **4/2 valves** are very often used to control double stroke cylinders.
- ❑ Internally, they are built as a combination of a 3/2 valve normally closed together with a 3/2 valve, normally opened.

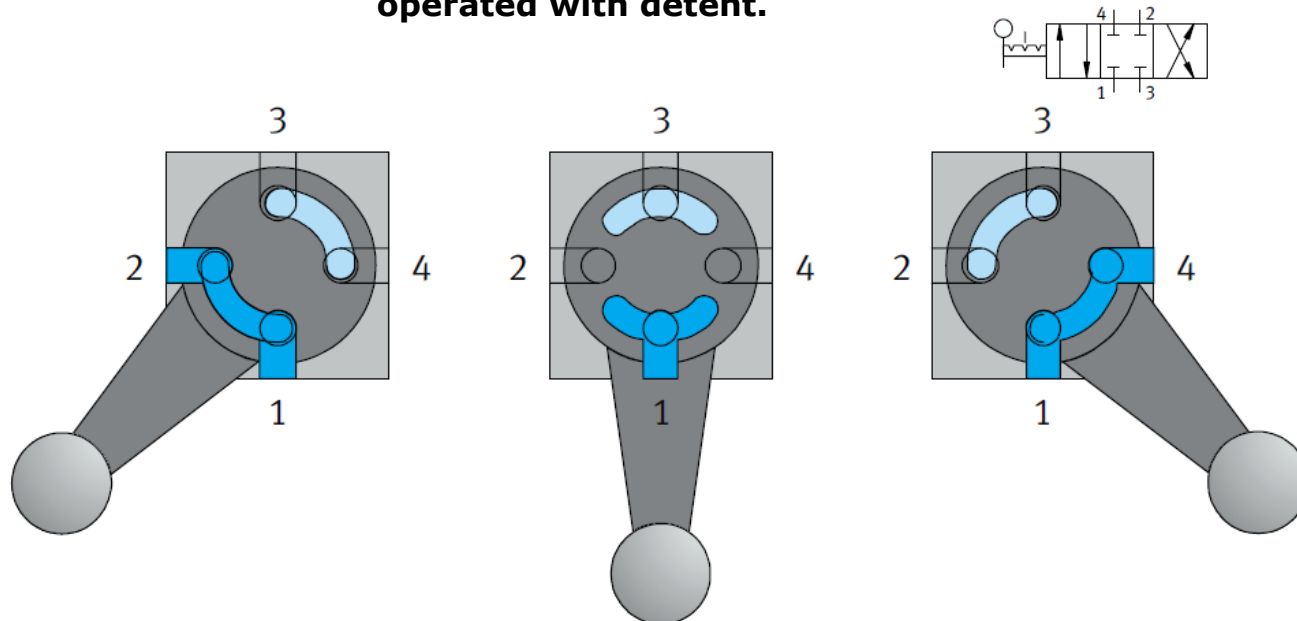




3.7. Pneumatic valves.

- ❑ The **4/3 valve** adds the possible load stoppage at the neutral position, besides the functions of the 4/2 valve.
- ❑ It is often used to include the possibility of cylinder blockage or stoppage, closing all possible flow paths. Pressure is released through the relief valve.
- ❑ They can have a manually operated spool with detent system (figure). By rotating the knob, the 3 positions are activated.

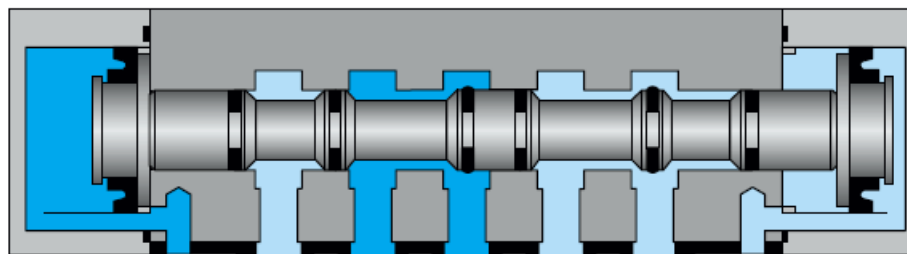
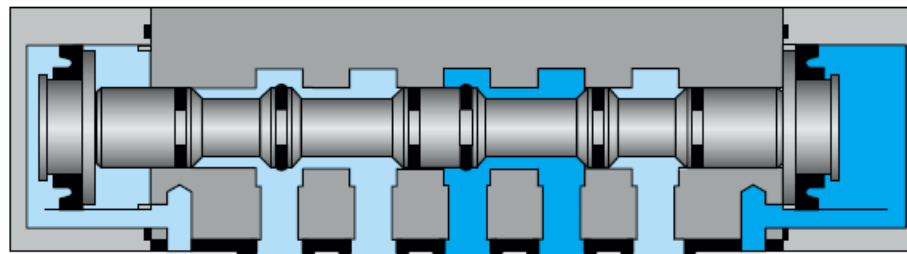
4/3 spool valve, with closed centre and manually operated with detent.



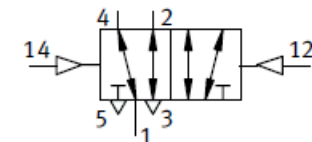
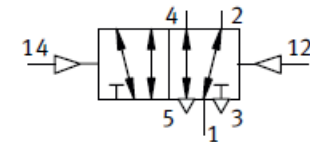


3.7. Pneumatic valves.

- ❑ A **5/2 valve** has the same functions of a 4/2 valve. It only differs in the use of the fifth way to make the exit of the air from cylinder independently.
- ❑ The **bi-stable impulse valve** has memory (a short pulse serves to switch). When there are signs in both connections, the first one dominates.



14 5 4 1 2 3 12



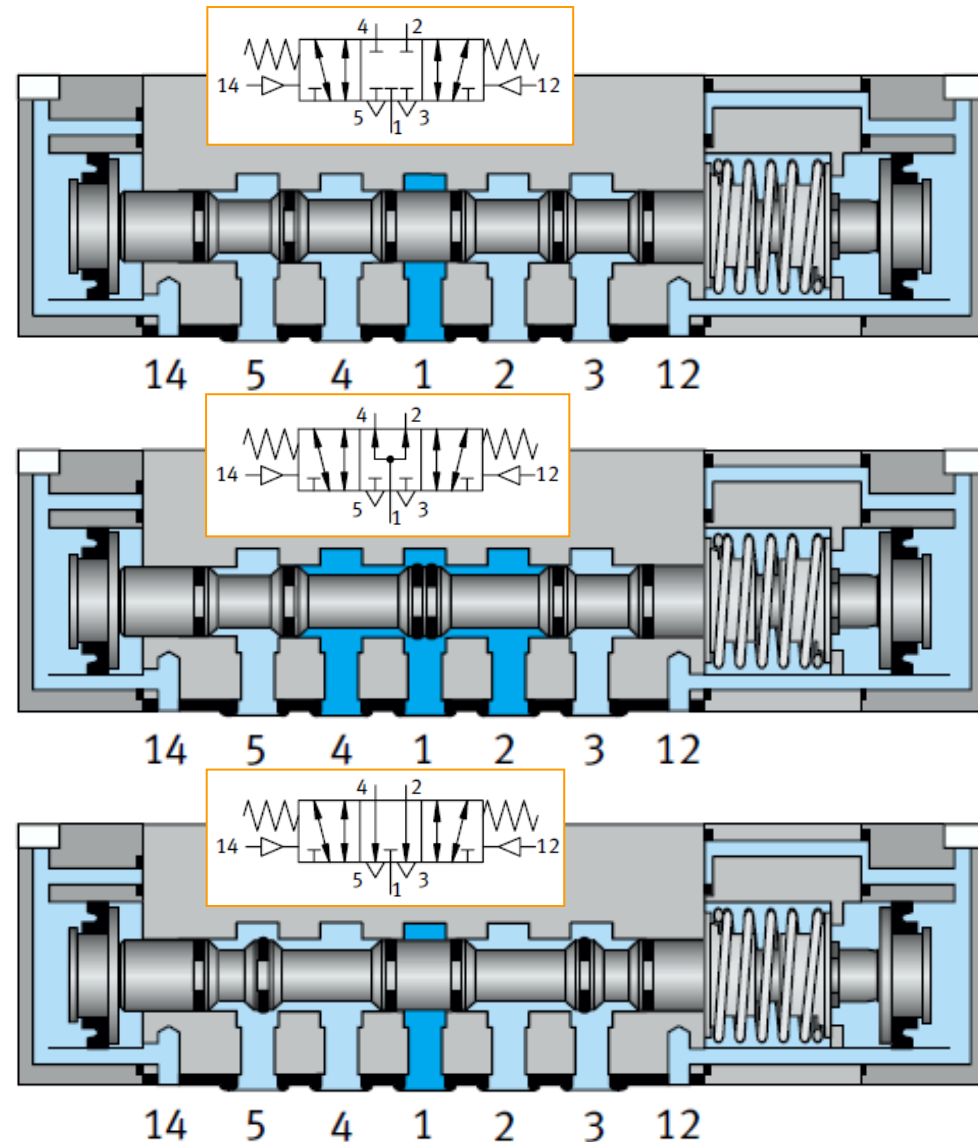


3.7. Pneumatic valves.

□ Besides of the previous functions, the **5/3 valve** has the added features of the central position. Usually this third position implies the cylinder stop and lock by any of its sides or to allow its free movement, without any pressure.

- Central position blocked.
- Central position with pressure.
- Central position to atmosphere.

□ If there is no pilot signal, the spring places the spool in the central position.

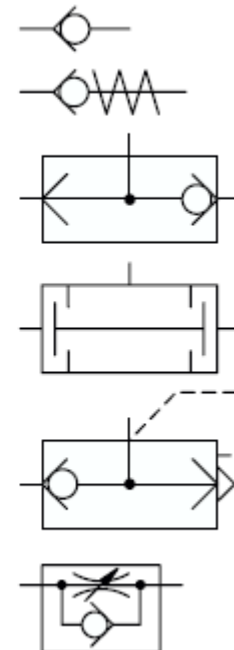




3.7. Pneumatic valves.

- ❑ The **shut-off valves** cut the flow pass in a given direction. Usually the block the flow pass in one direction and allow the free pass in another one. Often, they are built in a way that the pressurized air reinforces the shut-off effect (the air pressure imposes a force that works keeping the valve closed).
- ❑ Among these valves, the following can be found:

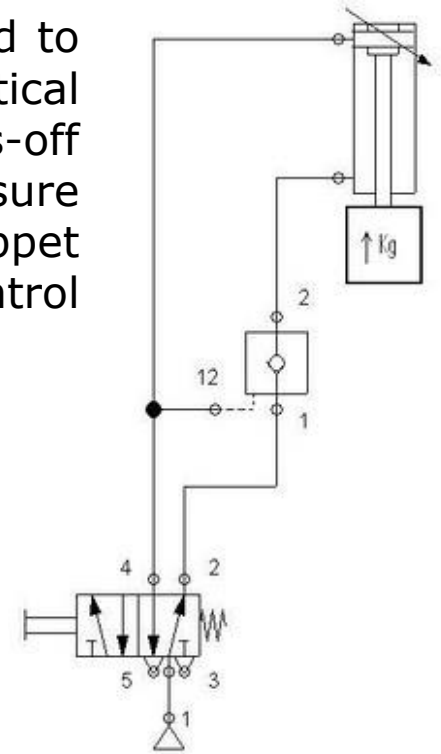
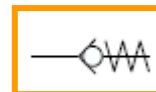
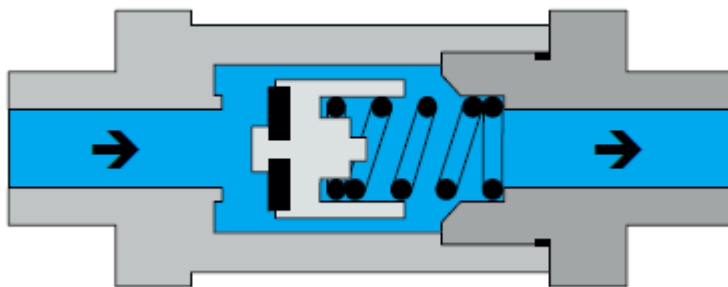
- Non-return (check) valve.
- Same but with spring opening.
- Circuit selection "OR" valve.
- Simultaneous signal "AND" valve.
- Quick exhaust (way out) for the fluid.
- Flow control in one direction.





3.7. Pneumatic valves.

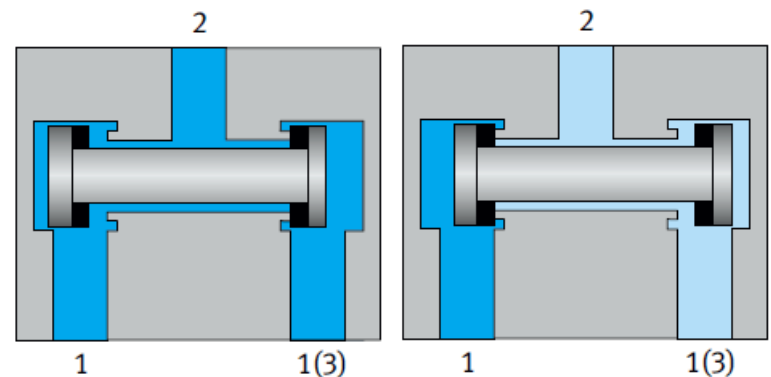
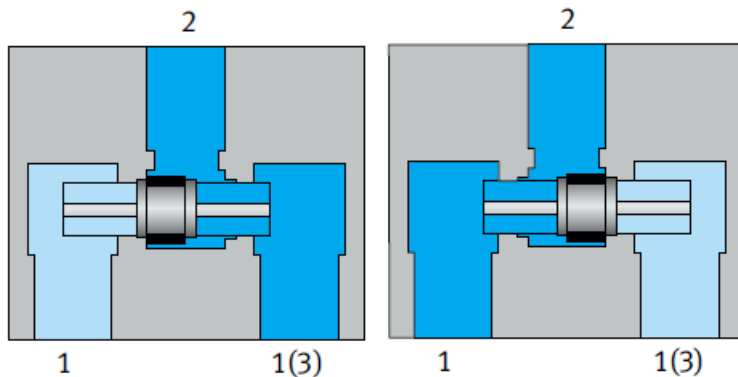
- ❑ The **non-return or check valve** lets the air pass in one direction and closes it in the opposite one. When the air pressure is higher than the spring force, the plate or poppet is moved and the air flows. In the opposite direction, the force is added to the spring force and the flow is blocked.
- It can be **piloted**. In such situations, it is often used to avoid a given movement of a double stroke vertical cylinder, with an external load, when the supply cuts-off or the compressor fails and, finally, there is a pressure drop in the system. A check valve with a plate or poppet provides more air-tightness than a directional control valve (often built with a spool).
- Example: **vertical cylinder**.





3.7. Pneumatic valves.

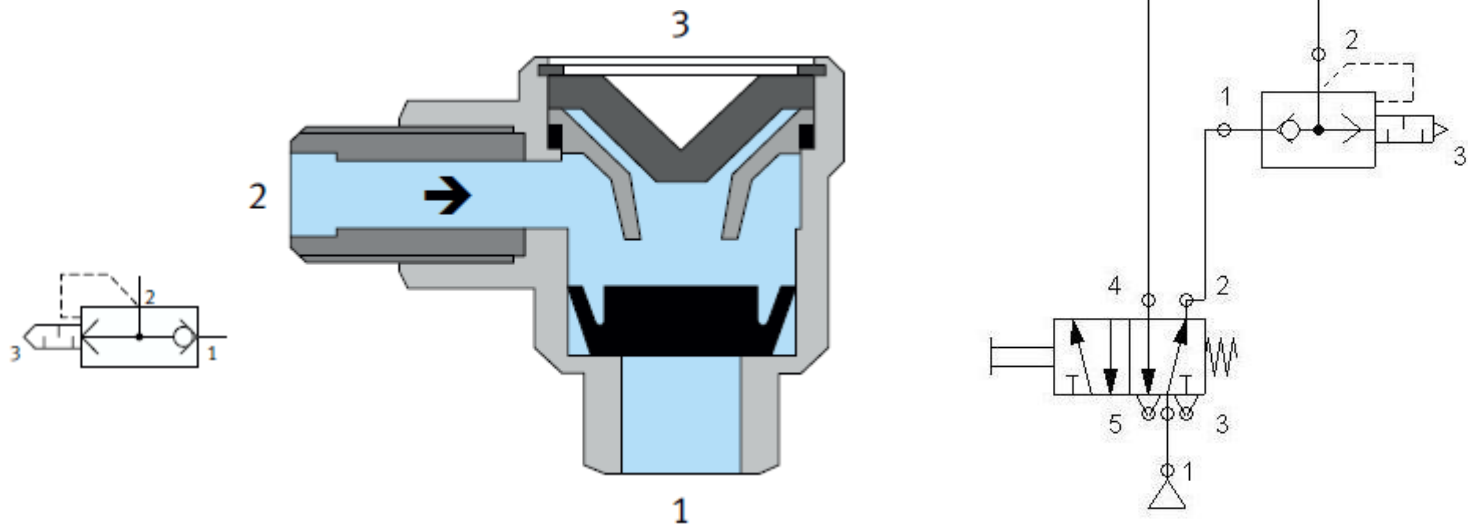
- ❑ The **logic valves “OR/AND”** allow to implement logic operations for the pass or blockage of a given pipe.
- The **selection valve (OR)** is used when the “OR” logical function is to be implemented. The air pressure signal at the valve inlets (1) produce a given output (2). In the case of no inlet signal, no output is obtained. When there is inlet pressures at both sides (not equal), the **maximum pressures** is passed to the output.
- The **simultaneously valve (AND)** is used to produce the “AND” logical function. The air pressure signals at the valve inlets (1) produce a given output (2) if both inlets have the same pressure signal. If different values are available for each input, **lower pressure** is passed to the output.

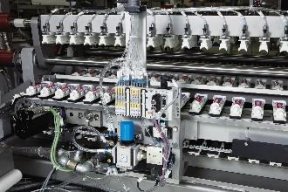




3.7. Pneumatic valves.

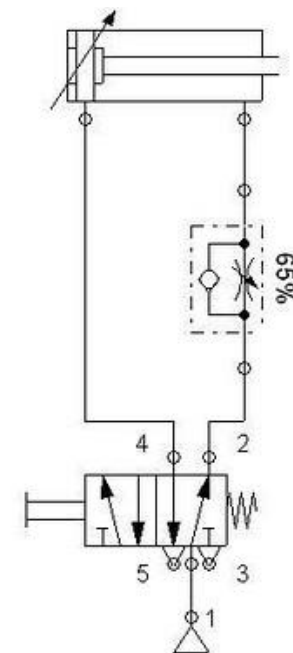
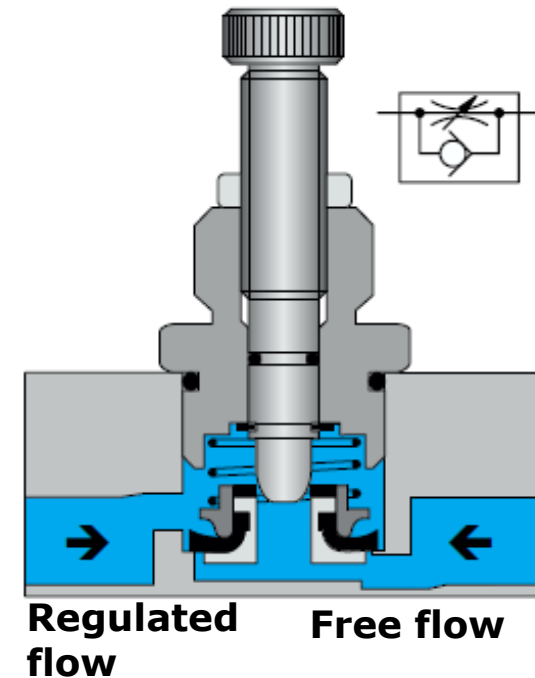
- ❑ The **quick exhaust valve** is used to quickly let the air out of the system. It means that, contrary to what happens in oleo-hydraulics, it is not necessary to have a return line, to bring back the fluid to the tank.
- ❑ The speed of the cylinder's movement can be raised in such way up to the maximum possible value, as in such movement, the air head losses are almost zero.
- ❑ They should be placed as close to the cylinder as possible.
- ❑ Example: **quick advance**.

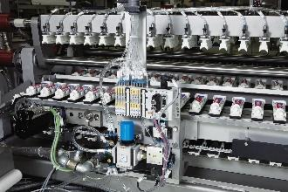




3.7. Pneumatic valves.

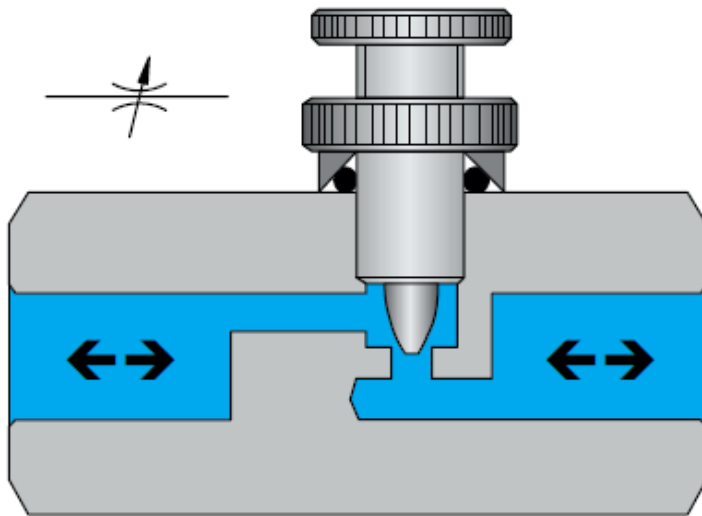
- ❑ The **non-return valve with flow control in parallel (flow control valve with by-pass in one direction)** is used to control the flow rate in one direction and let the flow pass freely in the opposite one.
- ❑ It is a very widespread valve to control the speed of the pneumatic actuators.
- ❑ To control the cylinder extent, it is always better to control the exhaust air in the rear chamber (rod side). To control the speed in the retract stroke, the opposite configuration is recommended. In this way, the cylinders are moved softly, without sudden jumps, which are the consequence of compression of the air and control the flow inside the same chamber where the air is directed. A softer and a more constant speed of the cylinder is obtained using this rule.



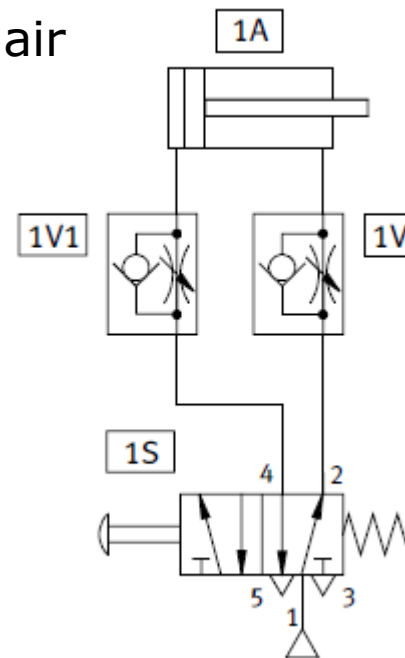


3.7. Pneumatic valves.

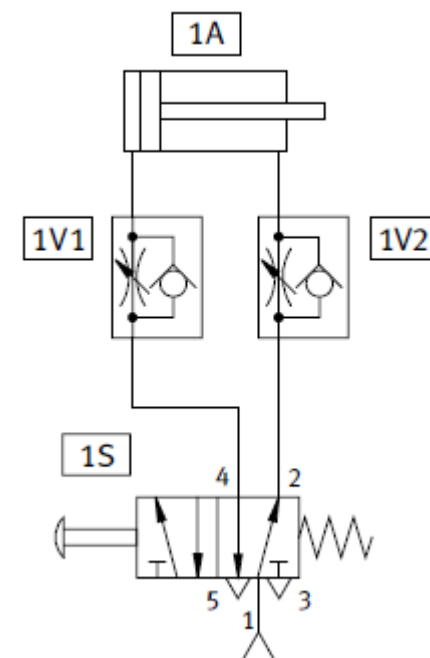
- ❑ The **flow control valves** allow a change in the compressed air that passes through them, and, therefore, control the speed of the cylinders or the velocity for a given sequence of actions in the system.
- ❑ It changes the pressurized air flow rate in both directions. Very often, these valves have variable control, by using an external screw, that changes the cross-section for the air flow.



Air flow control at the exit



Air flow control at the inlet

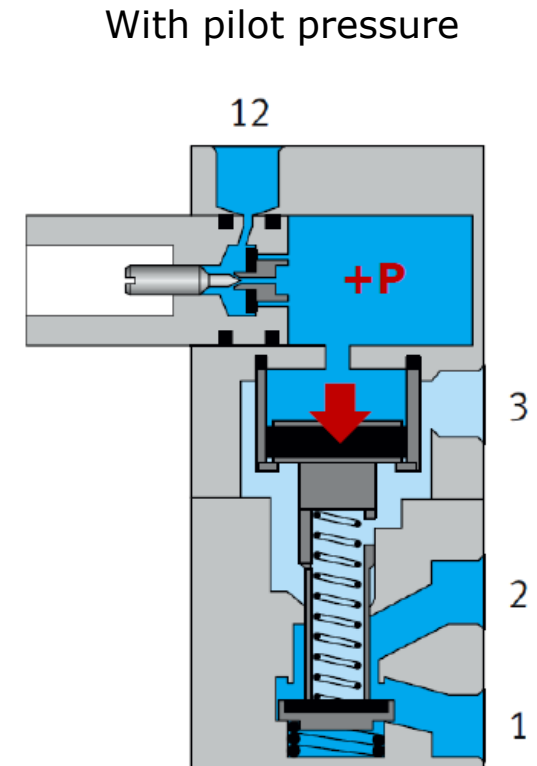
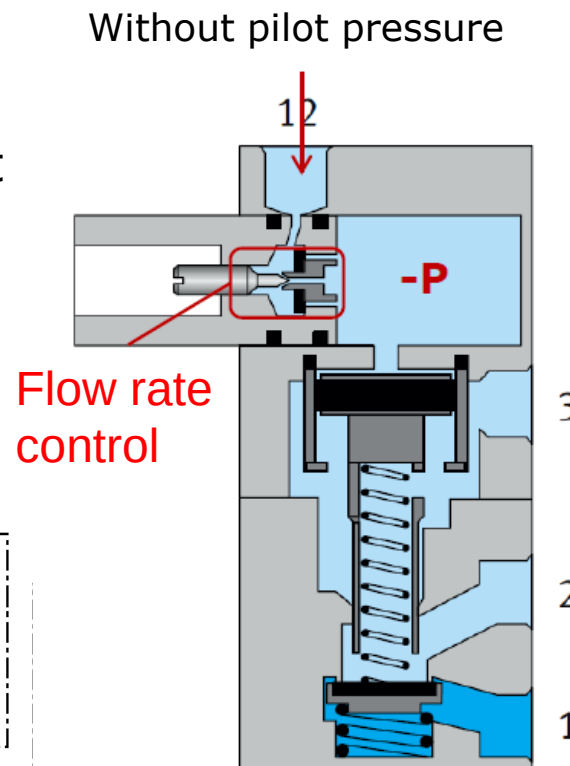
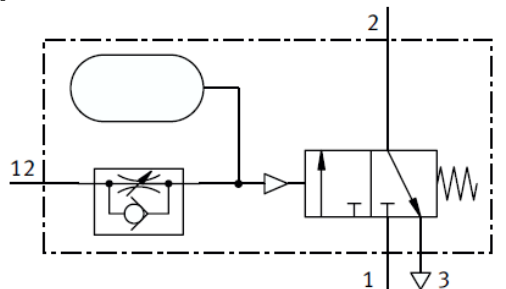




3.7. Pneumatic valves.

❑ A **pneumatic-timer valve** it is a pneumatic valve that results of the combination of two others. In particular, it is built with two valves and an air accumulator, as shown in the figure:

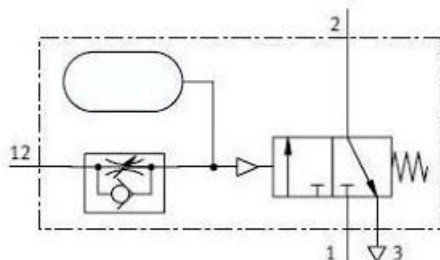
- A flow rate control with non-return in bypass.
- An air accumulator at pressure.
- A directional control valve 3/2, pneumatically operated.



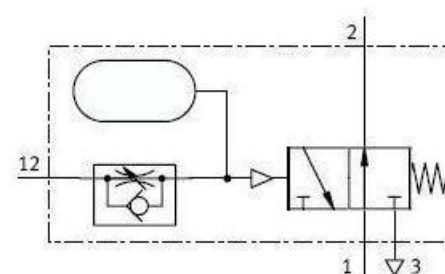


3.7. Pneumatic valves.

- ❑ The time control is done by limiting the flow rate through the line 12 to the accumulator. When the air inside this accumulator generates enough pressure to move the valve's spring, the directional valve is moved and allows the flow pass from 1 to 2.
- ❑ When line 12 is connected to the atmosphere, the flow leaves the accumulator through the non-return valve, without any flow control, allowing the quick closing of the directional valve.
- ❑ Depending on the positions of the directional valve (see the two figures), there is the possibility of a **timer with delay in the opening** or **with delay in the closing**...
- ❑ And depending of the 3/2 valve configuration, it could be **normally closed** or **normally opened**.



Timer with delay in the opening, normally closed.



Timer with delay in the closing, normally opened.

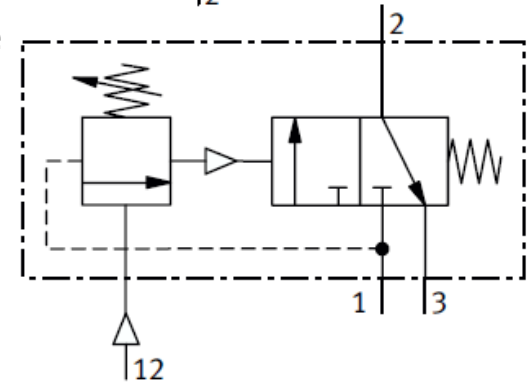
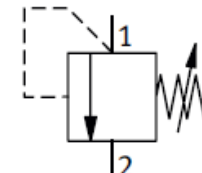
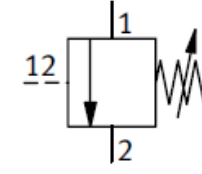
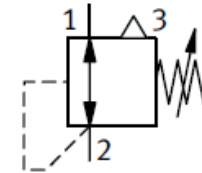
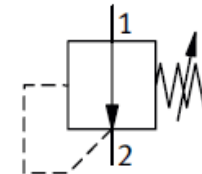


3.7. Pneumatic valves.

- ❑ The safe and efficient operation of a given pneumatic circuit needs the use of different means to control the pressure values. For such purpose, **pressure control valves** are used.
- ❑ There are many pressure control valves. The most usual ones are:

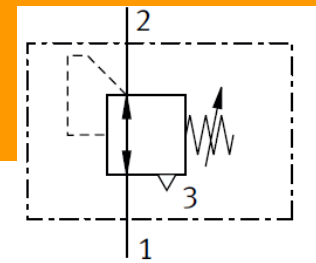
Pressure control valves:

- Pressure reduction valve without exit port.
- Pressure reduction valve with exit port.
- Pressure valve with pilot signal.
- Relief valve.
- Directional control valve together with pressure control valve.

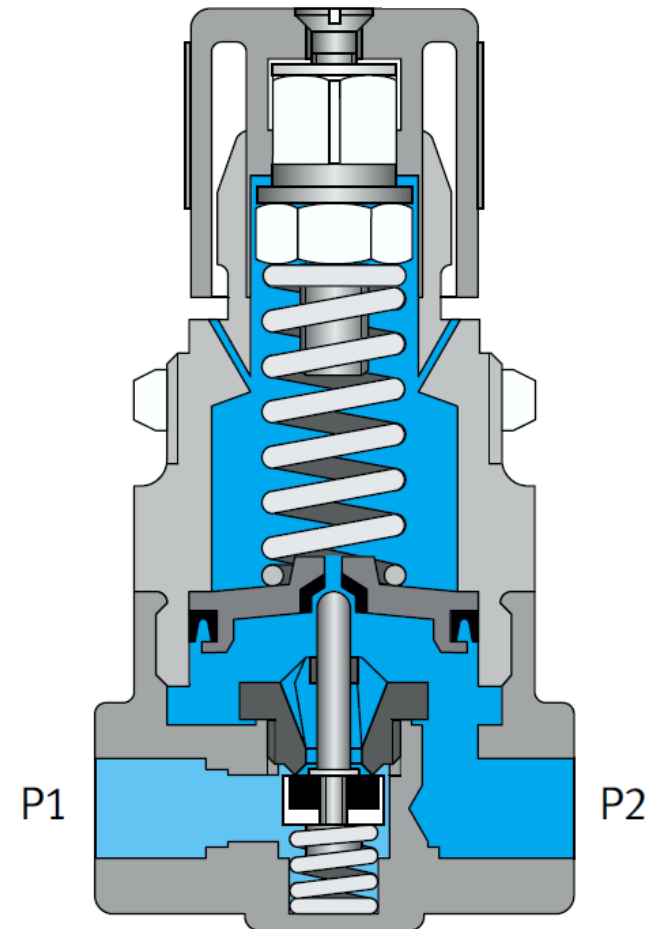




3.7. Pneumatic valves.



- ❑ The **pressure control valves** produce a constant pressure to the air downstream of their positions, if the compressor available pressure is higher than the useful needed one.
- ❑ The pressure control valve keeps an almost constant pressure regardless of the oscillations, delivery or consumption changes. Obviously, this does not work if the required pressure is higher than the available one.
- ❑ They can be with or without exit port.
- ❑ When they have **exit port**, if the required pressure decreases (for instance, when the required force in the cylinder lowers), the pressurized air is sent to the atmosphere through the exit port.



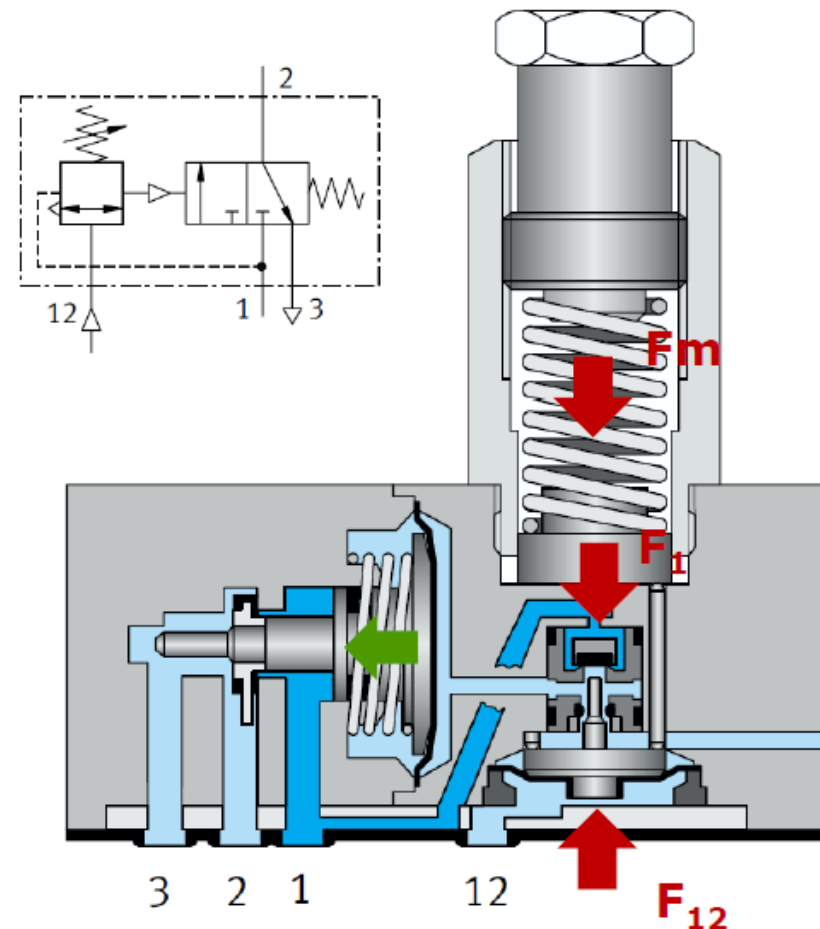
**Pressure control
valve with exit port**

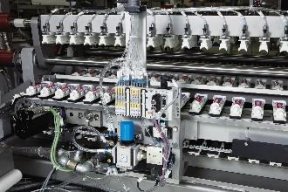


3.7. Pneumatic valves.

- ❑ The **sequence valves** have a similar working principle as the relief valves.
- They open the flow path when the pressure signal is higher than the one fixed by the externally regulated spring. The air flow from 1-inlet towards 2-outlet. The valve does not allow the flow pass until the control value in 12 is not reaching the reference value. When that value is reached, the inlet and outlet are connected (1-2). They are appropriate to commutation effects (pressure operated) as, for instance: a pressure signal when the handling force in a piece is reached.

**Pressure control valve
(sequence) combined with a
directional 3/2 valve**

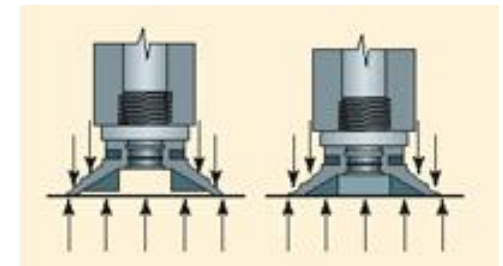




3.8. Vacuum.

Basic ideas.

- ❑ The **vacuum** could be defined as the lack or absence of air in a given volume. This air lack promotes a pressure decrease under the atmospheric pressure, reaching values close to the zero (absolute), promoting a pressure difference between the interior and the exterior of the volume casing.
- ❑ The external pressure produces a force on the casing that will increase for decreasing values of the interior pressure.
- ❑ An example of this behaviour is the operation of a vacuum cleaner. An under pressure is created in the pipe's end and air is sucked into the machine. In fact, the vacuum cleaned does not suck the air but promotes a pressure below atmospheric and then the air itself is conducted to the pipe due to the pressure difference.
- ❑ Another clear example is a suction pad, that is not supporting a given surface, but instead, is promoting a pressure below atmospheric and the surrounding external pressure is doing its job.

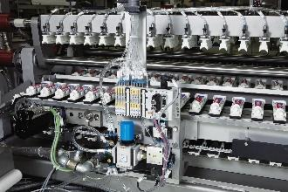


Forces in a suction pad



3.8. Vacuum.

- ❑ In this application, the flow goes in the “**opposite direction**” to what is the normal situation as the flow goes from the atmospheric zone (highest pressure) to the zones with lower pressure or vacuum pressure (lower pressure).
- ❑ The **pressure difference is quite limited**. It will never exceed a value of minus 1 atm.
- ❑ Nevertheless, it is not to be neglected the **flow capacity to produce a force** with an under-pressure. It is easily obtained the suction force of a surface equivalent to a paper page (20x30 cm²): $F = P_{\text{atm}} S = 1 \text{ kg/cm}^2 \cdot 600 \text{ cm}^2 = 600 \text{ kg}$ (more than half a ton).
- ❑ In the vacuum systems, some important features become critical, such as:
 - **Flow resistance** (the head losses should be as low as possible, as the pressure difference is not that high).
 - **Minimum volumes (avoid unnecessary volumes)** for the flow to move as quick as possible.
 - Two effects often become contradictories: lower head losses imply higher pipe diameters, but at the same time, this promotes the existence of higher volumes.



3.8. Vacuum.

- ❑ The energy consumption **increases asymptotically** (to infinite) when the vacuum gets closer to the absolute zero.
- ❑ a **vacuum degree** is obtained as a percentage of the atmospheric pressure finally obtained, for instance: 40.5 kPa (60%), 10.13 kPa (90%) or 1.013 kPa (99%)
- ❑ For a vacuum degree of 99%, a pressure of 0.01 atm has to be reached, meaning that the air volume should be 100 times higher than the initial one:

$$V_1 = V_0 \frac{P_0}{P_1} = \frac{1}{0.01} V_0 = 100 V_0$$

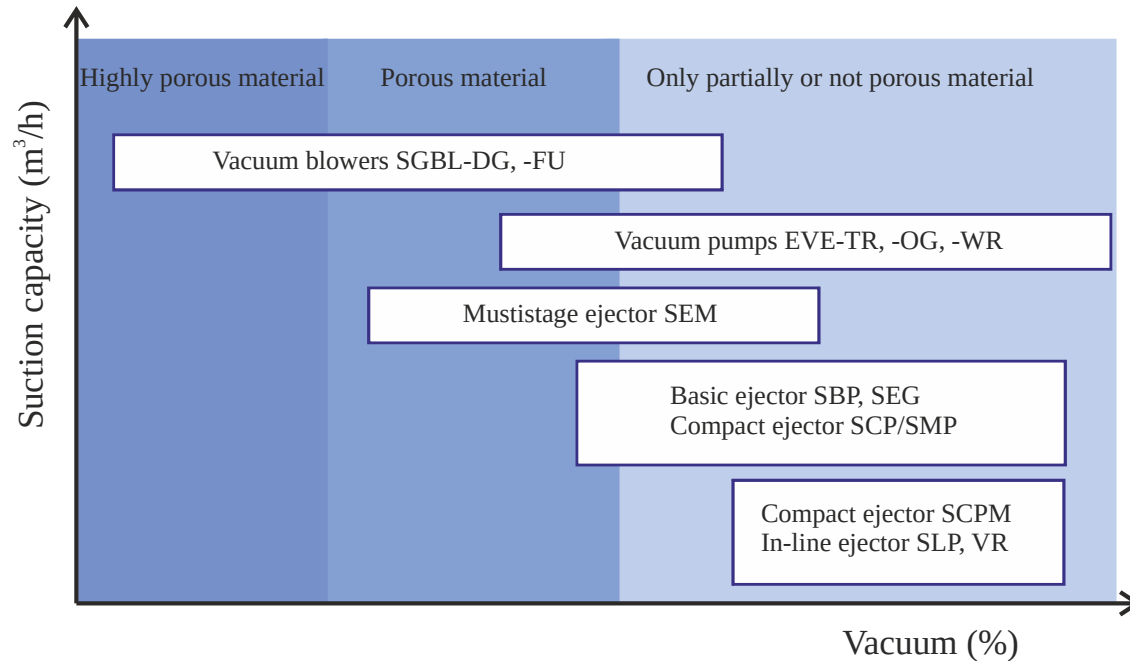
- ❑ Therefore, to optimize the energetic interchange, the **lowest possible degree** must be chosen. Depending on the practical application, there are typical ranges:

Low vacuum	Industrial vacuum	Process vacuum
0 – 20%	20 – 99%	> 99%
Cooling and handling.	Elevation, handling and automation.	Laboratories, micro-chips fabrication...



3.8. Vacuum.

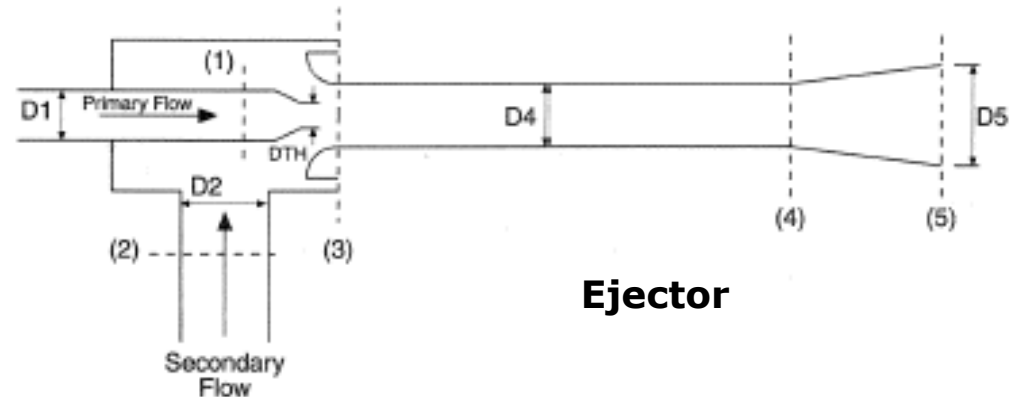
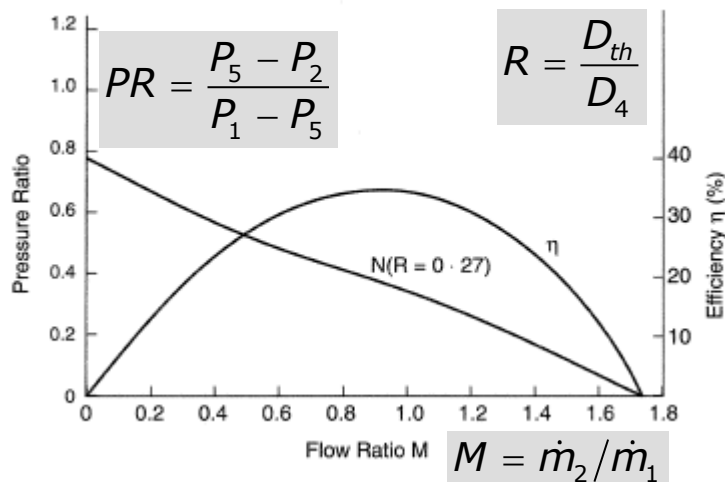
- ❑ The two most commonly used systems to produce a vacuum in industrial applications are **the mechanical pumping** (vacuum pumps) and the **ejectors** (based on the working principle of a Venturi).
- Among the mechanical pumping, different alternatives are possible. The most common ones are the blowers (centrifugal and channel ones) and the vacuum pumps (piston, membrane, vanes or lobe pumps).
- Considering the ejectors, there are single stage, two stages or, even, multi-stage units.





3.8. Vacuum.

- ❑ The vacuum pumps are, in a very simplified definition, compressors that work in an inverse mode (promoting a pressure under atmospheric).
- ❑ The ejectors use the kinetic flow energy to produce an under-pressure. This is done based on the Bernoulli's Principle:
 - A high speed jet (D_{th}), as part of the ejector, produces an under-pressure, that promotes the generation of a secondary flow. In section, D_4 , the mixture is produced. A pressure ratio (PR is defined).
 - They have an almost linear behaviour with the flow rate and depend on the ratio R , geometric relationship between the jet and mixture zone diameters.
 - For a gas-gas ejector, if $PR > 1.8$, there are some compressibility effects.

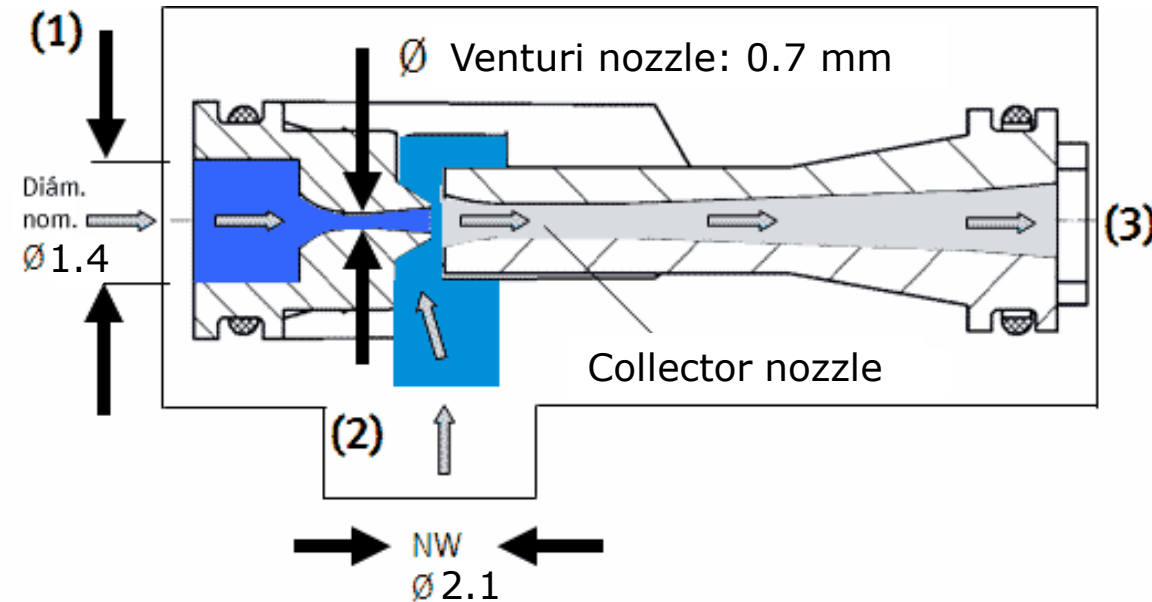


Ejector

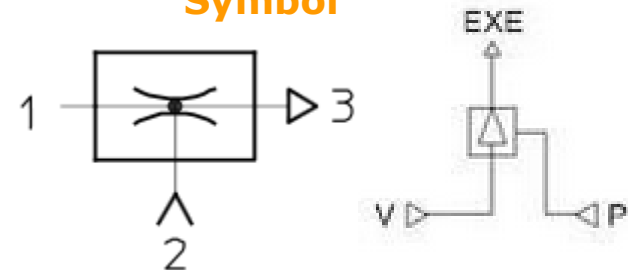


3.8. Vacuum.

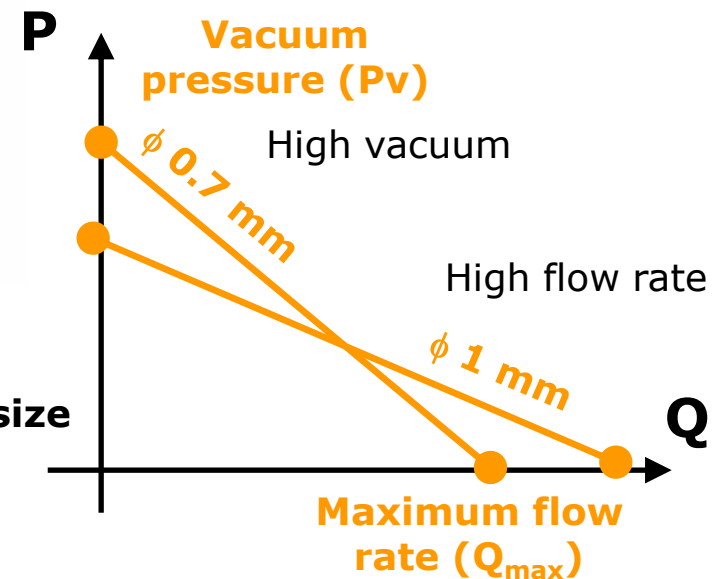
One stage ejector (T-type).

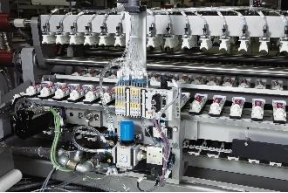


Symbol



Effect of the nozzle size on the ejector performance

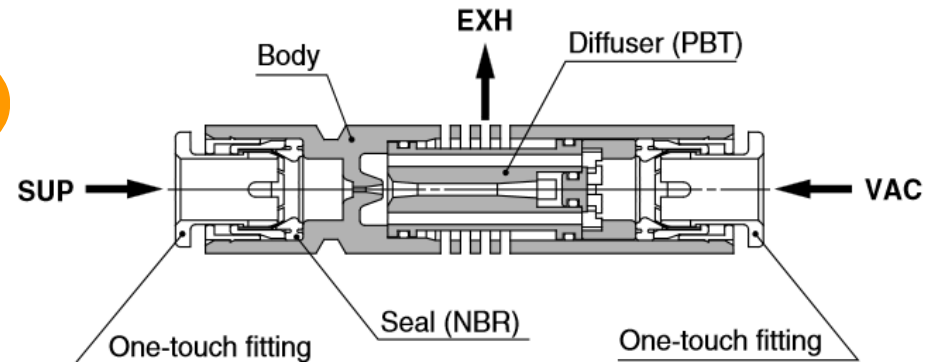




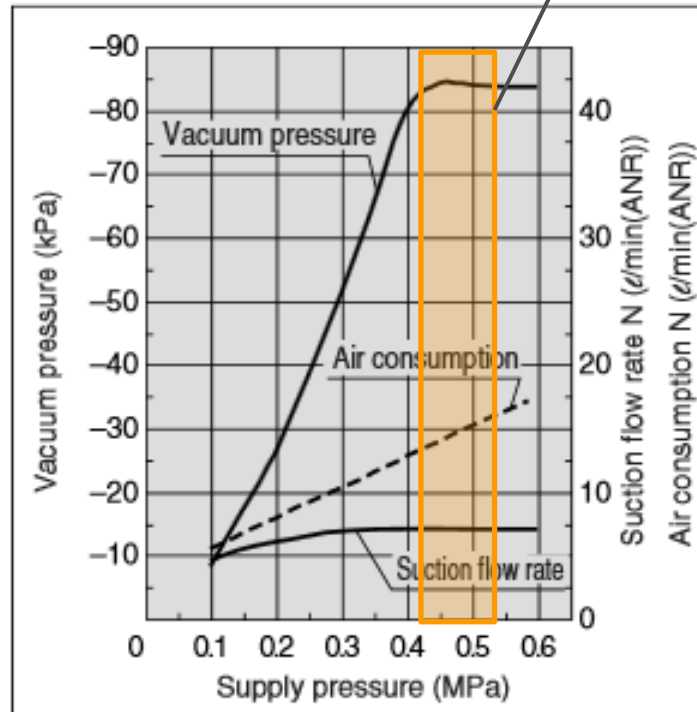
3.8. Vacuum.

One stage ejector (L type)

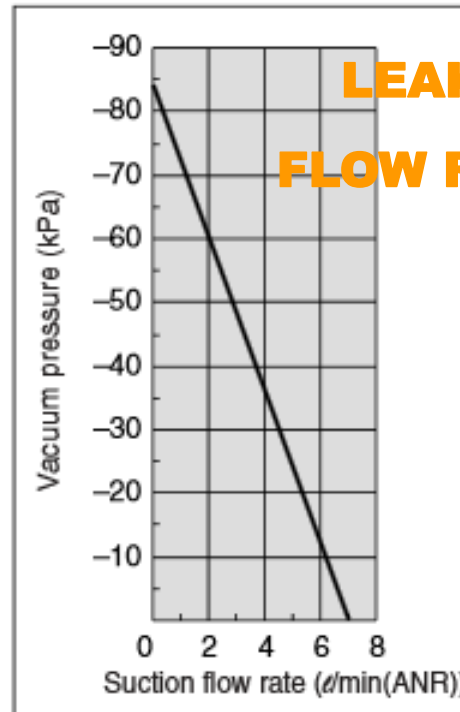
Better option for inlet pressure (4 – 4.5 bar)



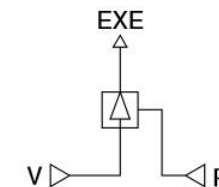
Exhaust Characteristics



Flow Characteristics



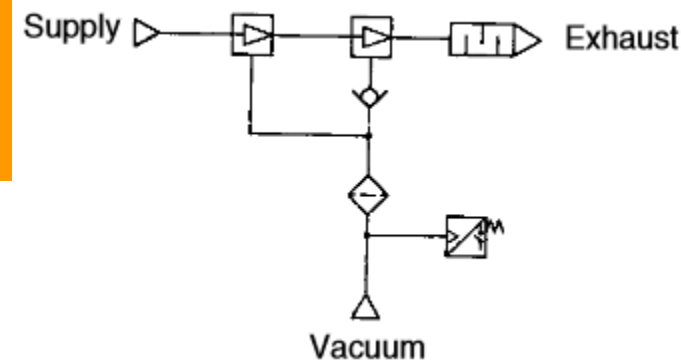
Circuit diagram





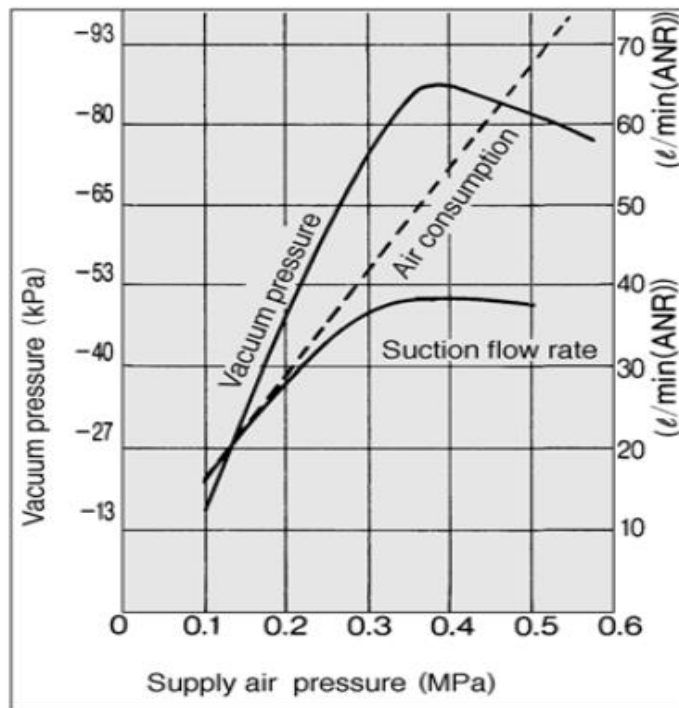
3.8. Vacuum.

Two stage ejector.

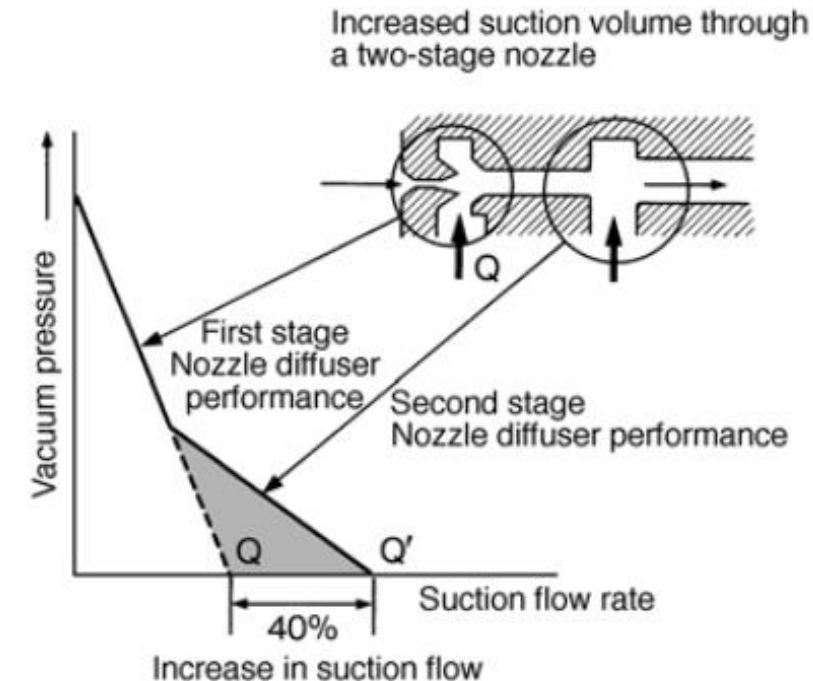
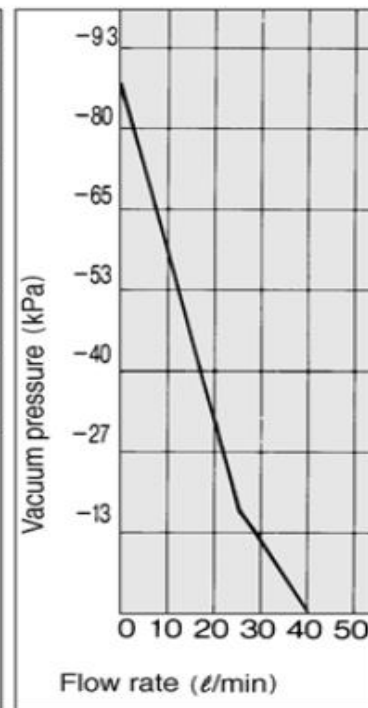


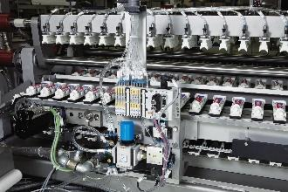
- The first stage provides a strong vacuum.
- The second stage with a high suction flow rate at the beginning, is then bypassed with a non-return valve, when it reaches the maximum vacuum level for the stage.

Exhaust characteristics



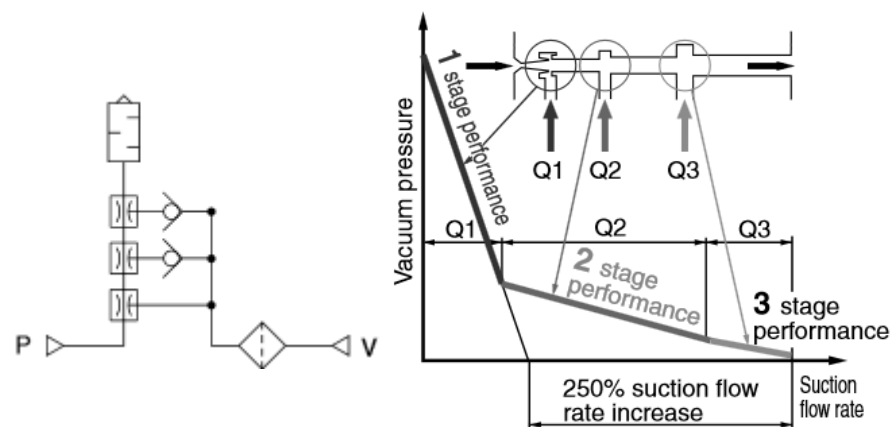
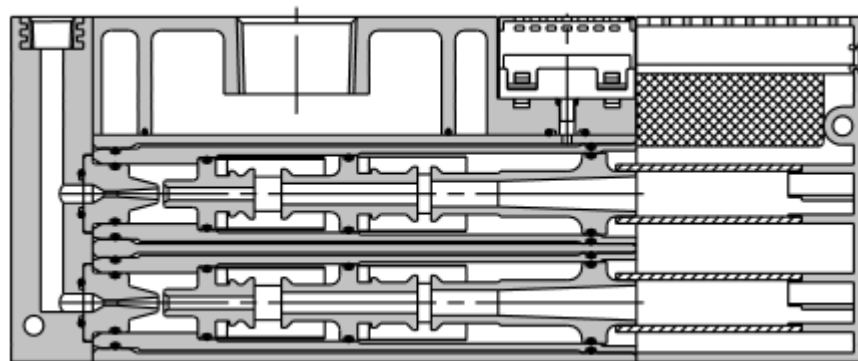
Flow characteristics



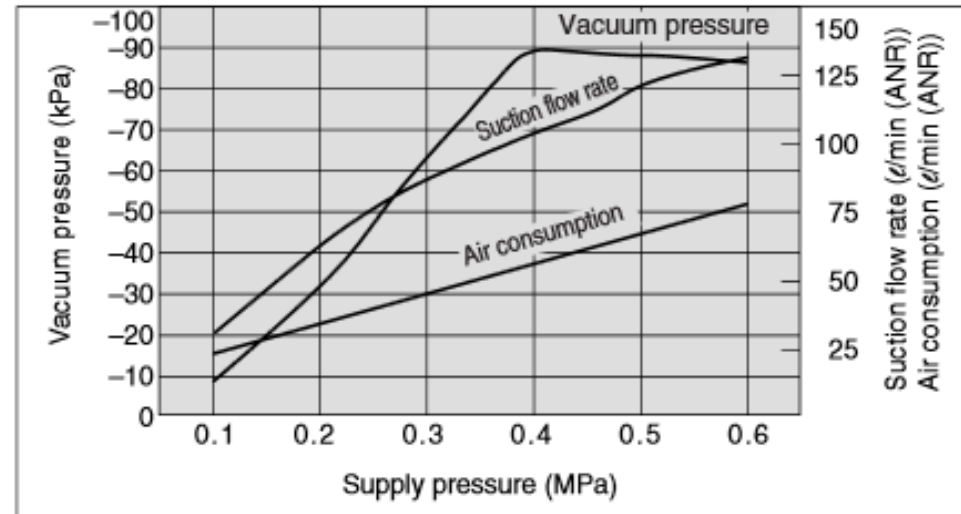


3.8. Vacuum.

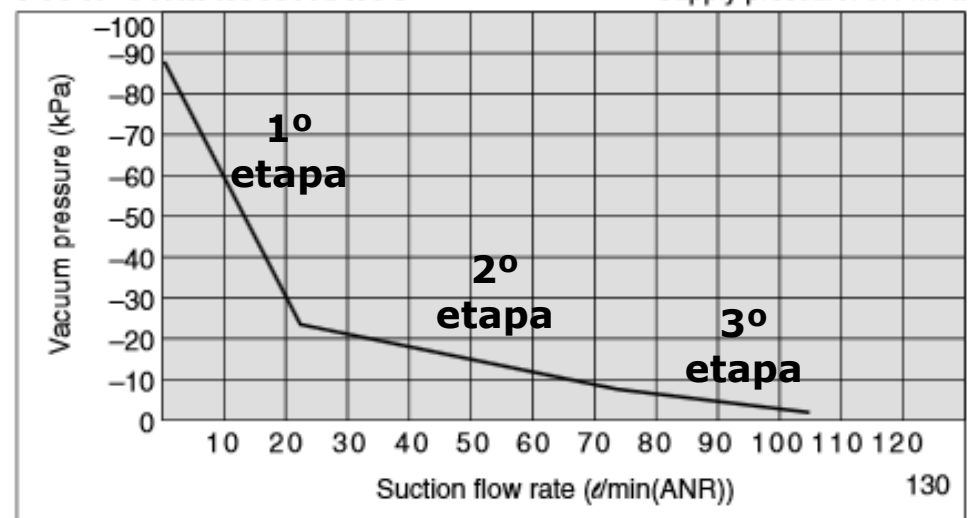
Multi-stage ejector (three stages)



Exhaust Characteristics



Flow Characteristics



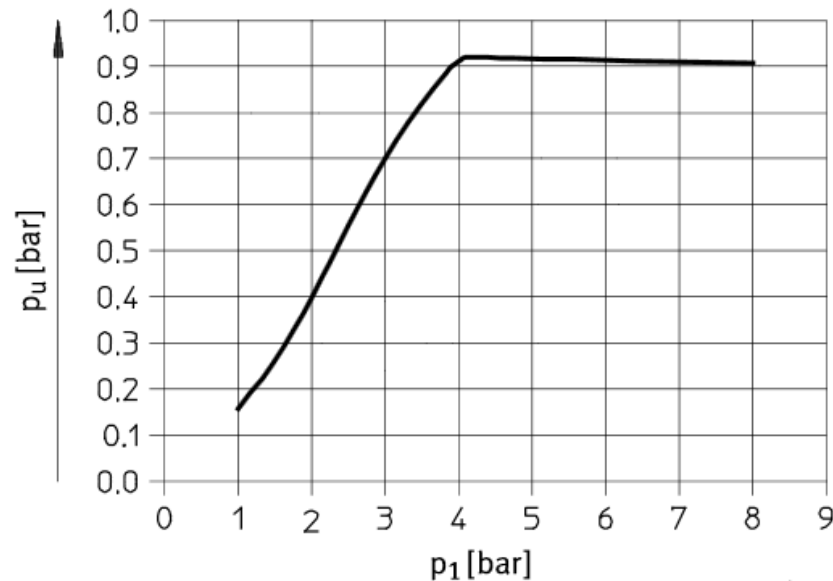


3.8. Vacuum.

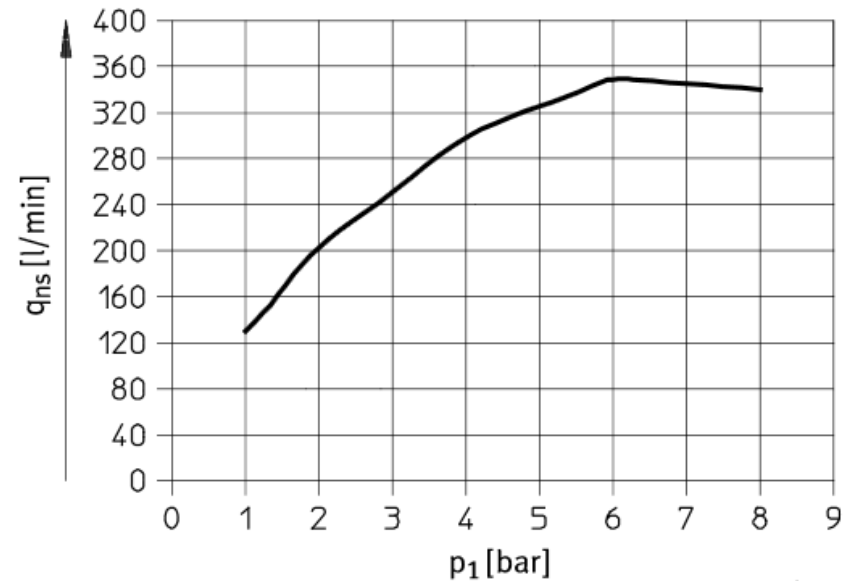
Ejector type selection.

- There are two possibilities: ejector promoting an **strong vacuum** (up to 93%) or an ejector promoting a **high suction flow rate** (up to 339 l/min). That is, a high suction pressure or a high suction flow rate.
- It depends on the pressure of the delivery compressor available, p_1 .

Vacuum p_u as a function of operating pressure p_1



Suction rate q_{ns} as a function of operating pressure p_1

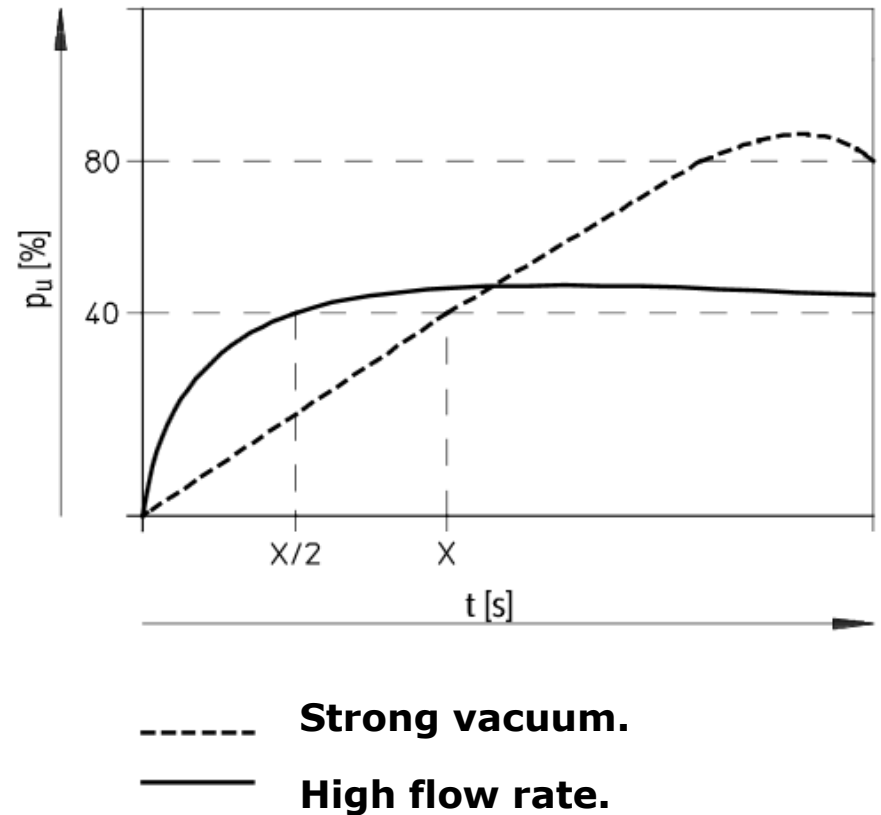


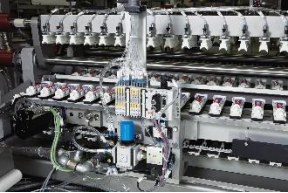


3.8. Vacuum.

Ejector type selection: comparison.

- ❑ An ejector for strong vacuum or suction is optimized to create quite low pressures (< -0.4 bar, relative), and is suitable for most of the usual industrial applications.
- ❑ A high flow rate ejector is optimized for a high flow rate suction at relative high pressures (not lower than -0.4 bar, relative). It would better work for applications with **important flow rate leakages**.





3.8. Vacuum.

Ejector type selection: leaks.

- ❑ Depending on the object to be moved (shape, porosity, surface quality), there will be air tightness or leakages between the object and the suction pad. Equivalently, there could provide air tightness in the pipes up to the ejector. In general, two possible types are possible:
 - Airtight systems.
 - Systems with leakages.
- ❑ In the **airtight systems** it is considered that the vacuum degree obtained is the maximum that the ejector may provide. Therefore, the flow capacity of the ejector will only depend on the air volume to be delivered and the answering time required:

$$Q_1 = V \frac{60}{t_1} [NL/min]$$

$$V = \frac{\pi D^2 L}{4 \cdot 1000} [L]$$

$$Q_{\max} = (2 \div 3) Q_1$$

$$t_2 = 3 t_1$$

D = Pipe interior diameter (mm).

L = Pipe length (m).

V = volume to be delivered (litres).

Q_1 = average required flow rate (NL/min).

t_1 = answer time for 63% of Vacuum pressure (s).

t_2 = answer time for 95% of Vacuum pressure (s).

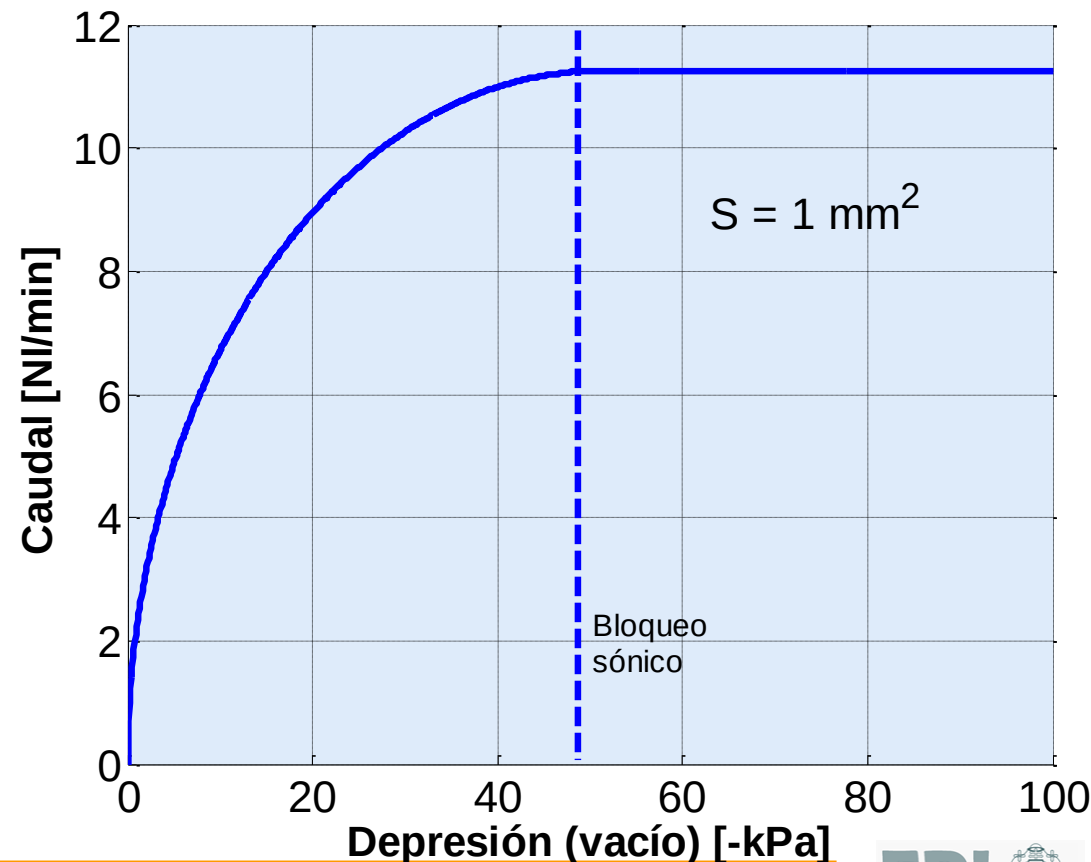
$Q_{\max}$ = maximum flow rate needed in the ejector.



3.8. Vacuum.

Ejector type selection: leaks.

- ❑ In the **systems with leakages** it is necessary to evaluate these leaks to be able to compensate for them with an ejector with a suction capacity higher than the air-tightness system.
- ❑ If the value of the section that provides the required head loss is known, it is quite simple to determine the leakage flow, using the graph in the right (same equations as in slide 17, part I of Chapter 3).
- ❑ For sections different to 1 mm^2 , it is needed to multiply the value in the graph by the available value.



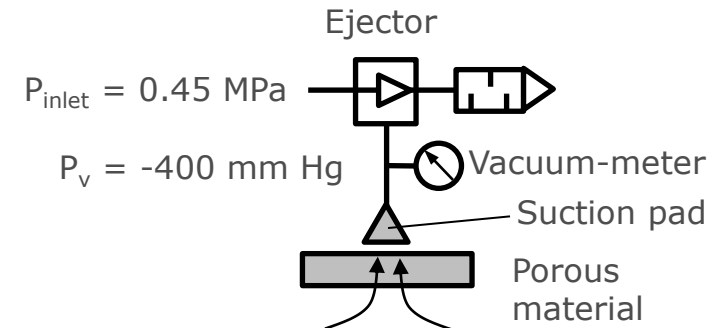


3.8. Vacuum.

Ejector type selection: leaks.

- ❑ If the leakage due to the suction is unknown (usual situation), an analogy with a simplified sketch may be used.

- **Example:** a suction pad to move a porous material is being built. The ejector has an inlet pipe with air at 4.5 bar. When the vacuum pressure is stable, the vacuum-meter reads -400 mmHg. A single stage L-type ejector is used (see slide 58). What is the leakage flow rate, Q_L ?



- **Solution:** With inlet pressure of 4.5 bar, the ejector provides -85 kPa and produces a suction flow rate equal to 7.5 NI/min when it works in an airtight system. But the vacuum is only -400 mmHg, that is, -53.3 kPa. Using the graph with the characteristics, for such pressure, a value of 2.5 NI/min is found. Meaning that the flow leakage is $Q_L = 2.5 \text{ NI/min}$.



3.8. Vacuum.

Ejector size calculations.

- For an airtight system, the flow rate at the ejector has been obtained in slide 63. The value for $Q_{\max}$ is taken as the necessary to define the size and diameter of the ejector.
- In the system with leakages, the same equations as for the airtight system are used, but adding an additional flow rate Q_L to the already required Q_1 :

$$Q_1 = V \frac{60}{t_1} + Q_L \text{ [NL/min]}$$

- The other parameters are obtained also, according to:

$$V = \frac{\pi D^2 L}{4 \cdot 1000} \text{ [L]}$$

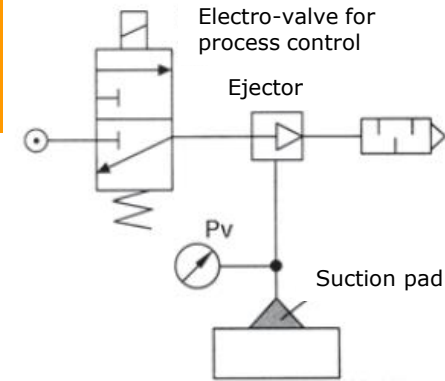
$$Q_{\max} = (2 \div 3) Q_1$$

$$t_2 = 3 t_1$$

- **MIND:** The times t_1 and t_2 are still to be defined and are related to the ejector answering times. **But then, how are the answering times defined?**



3.8. Vacuum.

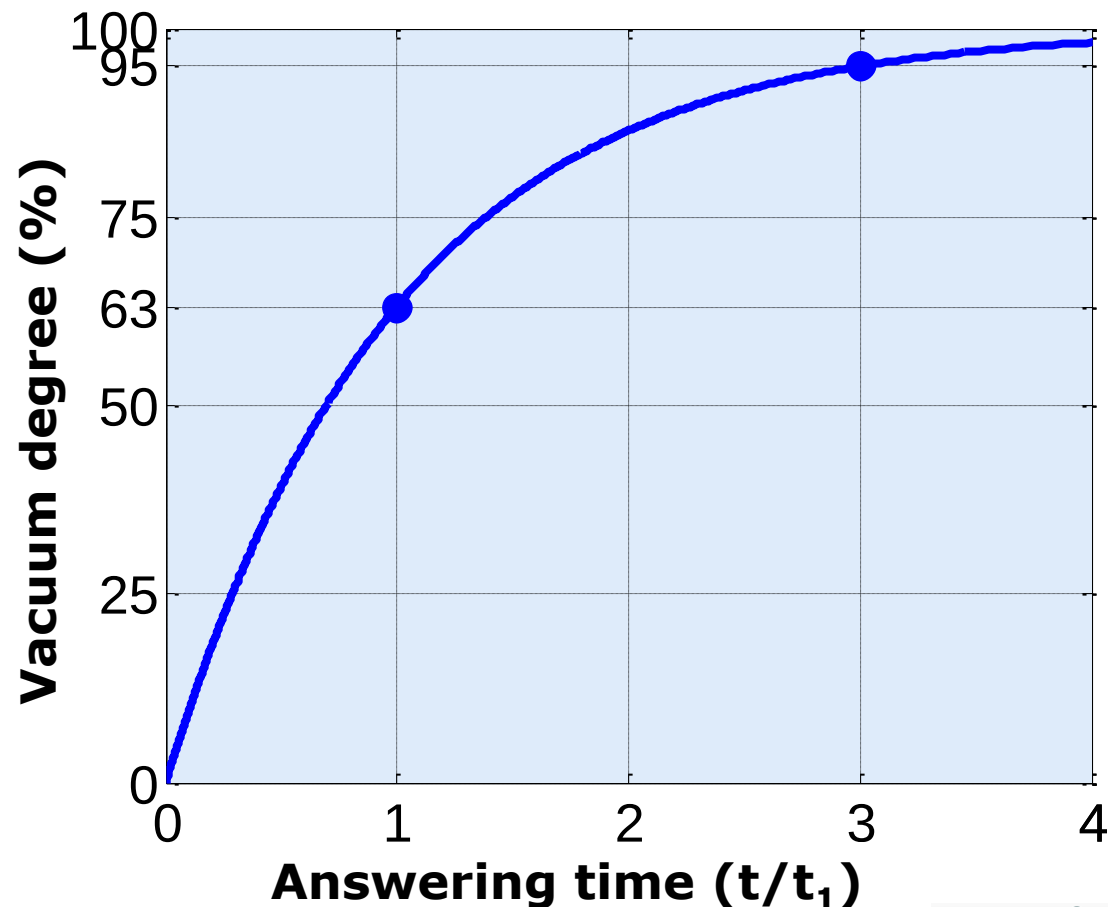


Answering time calculation.

- Once the flow is established in the ejector, an answering time is required, with exponential distribution:

$$t = \underbrace{\frac{V}{Q}}_{t_1} \ln \left(\frac{p_{atm}}{p_{vac}} \right)$$

- After the opening step, the system answers as drawn in the graph, where t_1 and $t_2 = 3t_1$ are the reference values.



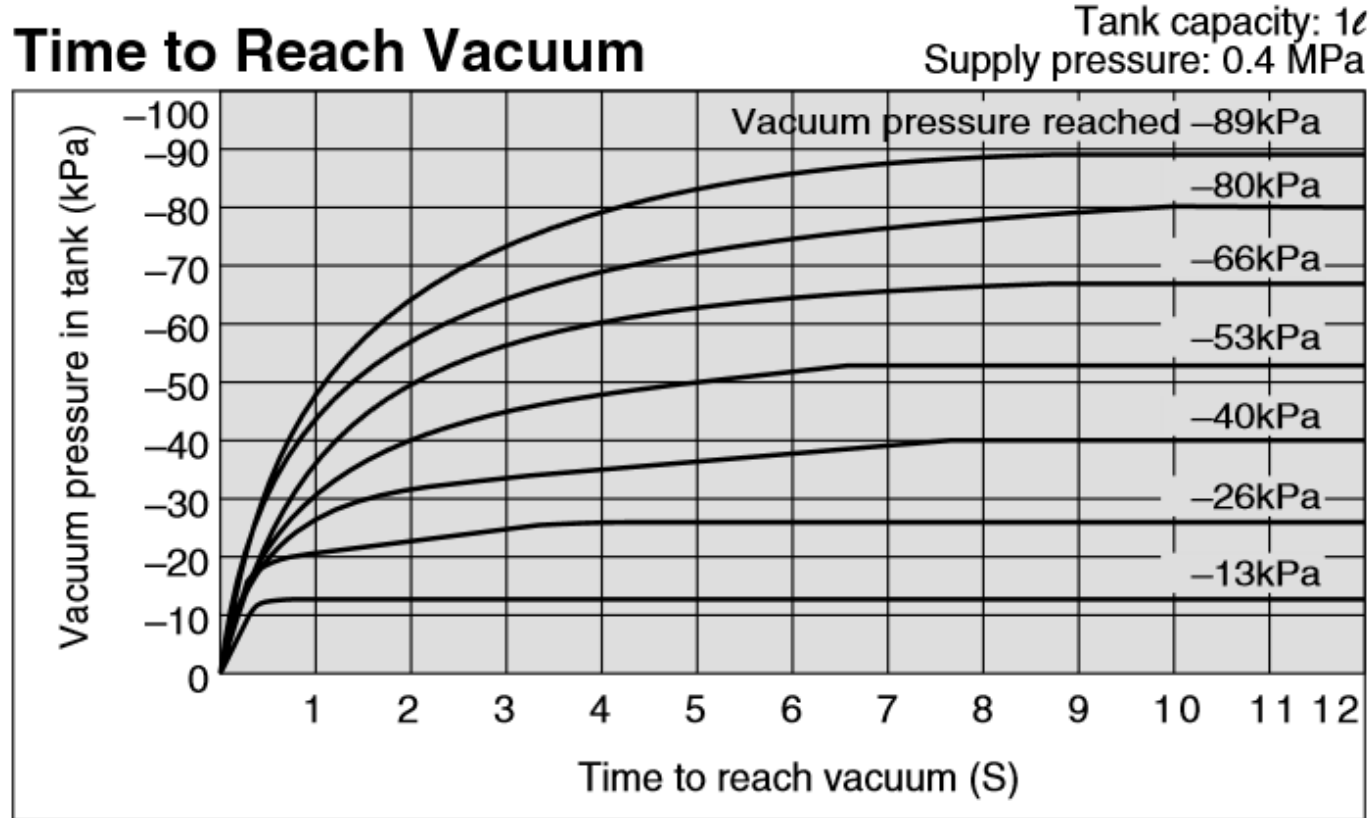


3.8. Vacuum.

Answering time calculation.

- ❑ The manufacturers provide graphs of answering times versus the vacuum pressure, the pressure at the inlet and the volume to be delivered (see graph below).

Time to Reach Vacuum



➤ Also, analytically:

$$t_1 = V \left(\frac{60}{Q} \right) [s]$$

$$t_2 = 3 t_1 (s)$$

As the value for Q , the lower among Q_1 and Q_2 , obtained as:

$$Q_1 = \left(\frac{1}{2} \div \frac{1}{3} \right) Q_{\max}$$

$$Q_2 = 11.1 S$$

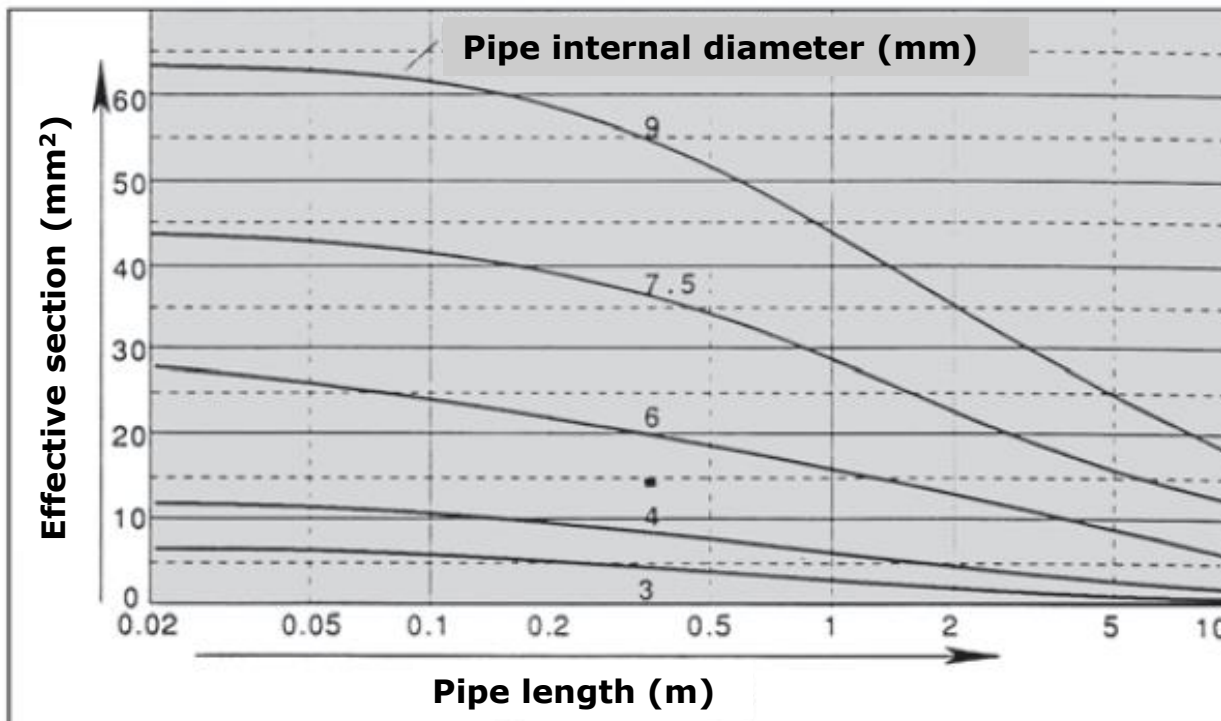
$$V = \frac{\pi D^2 L}{4 \cdot 1000} [L]$$



3.8. Vacuum.

Volume to be delivered.

- It is required to know the volume to be delivered by the pipes. The good practice indicates that the pipes should be smooth and as low head loss as possible is required, and also the lower the volume is, the better.
- The volume of suction pads and other elements must be added to the calculation.



➤ This graph allows the simple calculation of the optimum value for a given length and pipe diameter. It includes the calculation:

$$S = \alpha \sqrt{\frac{D^5}{L}} \text{ (mm}^2\text{)}$$

$$\alpha = 0.0844 C_t D^{0.155}$$

$C_t = 1.6$ – iron

$C_t = 2$ – plastic/cooper

➤ And for a $Q_{\max}$ and the unknowns are L and S , then:

$$S = \frac{Q_{\max}}{L} \text{ (mm}^2\text{)}$$



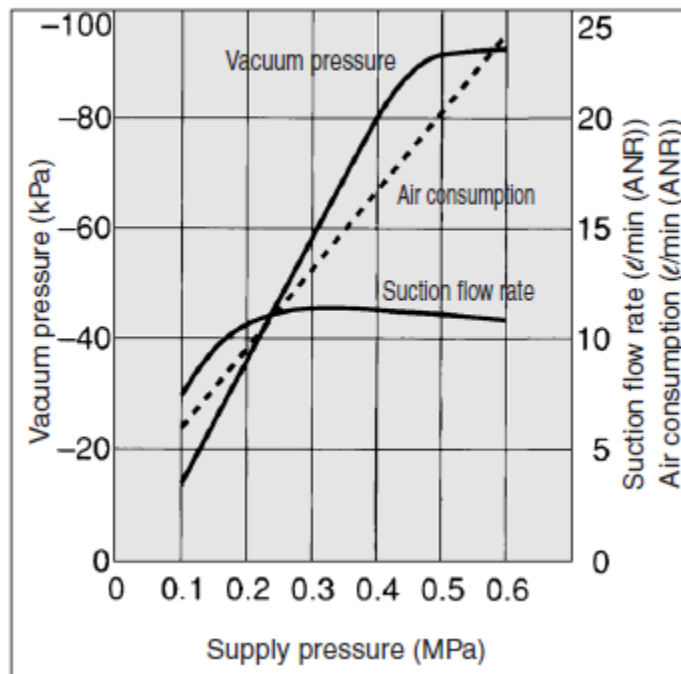
3.8. Vacuum.

Example:

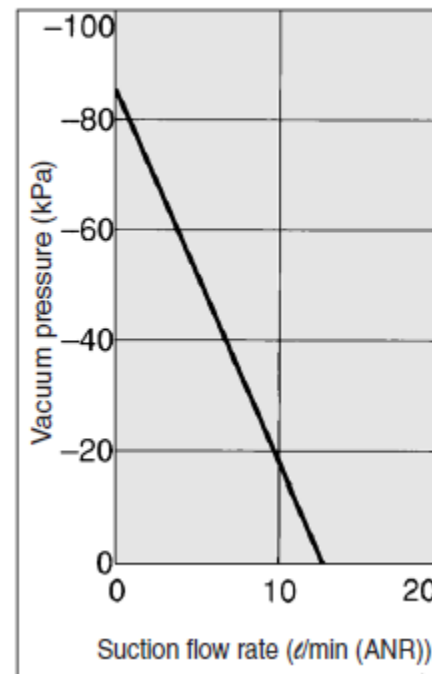
- ❑ Ejector type ZH07-S. Inlet pressure 0.4 MPa.
- ❑ Pipe length 1m, with pipe internal diameter, 6 mm.
- ❑ Suction pad diameter, 10 mm.
- ❑ Case a) Air tightness. Case b) Final vacuum pressure: -65 kPa.

ZH07-S

Exhaust Characteristics



Flow Characteristics





3.8. Vacuum.

Solution – case a)

- ❑ As a first step, in the graphs, it is found that for an inlet pressure of 0.4 MPa, the maximum vacuum pressure would be -80 kPa. And, the maximum suction flow rate would be 12 l/min.
- ❑ The average suction flow rate is, then: $Q_1 = \left(\frac{1}{2} \div \frac{1}{3}\right) Q_{\max} \approx \frac{1}{2.5} 12 = 4.8 \text{ NI/min}$
- ❑ Now, for a 1 m length (in the pipe), it is obtained:
$$S = Q_{\max}/L \approx 12/1 = 12 \text{ mm}^2$$
- ❑ Therefore, it is found Q_2 as: $(11.1) S = (11.1) 12 = 133.2 \text{ NI/min}$.
- ❑ The volume to be delivered is: $V = \pi D^2 L / 4000 = \pi \cdot 6^2 \cdot 1 / 4000 = 0.028 \text{ L}$
- ❑ The volume of the suction pad could be added, but it is really negligible:
$$g \approx \pi D^3 / 6 = \pi 1^3 / 6000 = 0.00052 \text{ l}$$
- ❑ Calculating the time t_1 as $t_1 = 60 g / Q_1 = 60 (0.028 / 4.8) = 0.35 \text{ s}$, which is the answering time to reach the 63% of the vacuum pressure. As a matter of fact, $t_2 = 3 t_1 = 1.05 \text{ s}$ to reach the 95% of the vacuum pressure.



3.8. Vacuum.

Solution – case b)

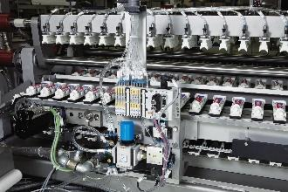
- ❑ Let us determine the leakage flow rate. As it was found before, with an inlet of 0.4 MPa, the vacuum pressure was -80 kPa, with a maximum flow rate of 12 NI/min. If the final pressure in the system is -65 kPa, then, the losses are 3.5 NI/min.
- ❑ For the losses or air leakage calculation, the average flow rate is $Q_1 = 4.8 - 3.5 = 1.3$ NI/min, and then: $Q_{\max} = 2.5 Q_1 = 2.5 \cdot 1.3 = 3.25$ NI/min
 ↙ (Theoretical value)
- ❑ Q_2 is determined as previously, that is: $(11.1) S = (11.1) 12 = 133.2$ NI/min and the volume to be delivered would still be 0.028 l. Finally, the answering time is obtained as: $t_1 = 60 V / Q_{1E} = 60 \cdot 0.028 / 1.3 = 1.29$ s
- ❑ The time for the 95% vacuum will be then: 3.87 s.
- ❑ If we would be finding the dimensions, the air leakages should be added to be able to get o a reasonable suction, and then, a bigger ejector should also be chosen.
- ❑ **MIND:** The air volumes in the suction pads up to 50 mm in diameter, can be neglected. From that value, they are often (in industry) considered as an additional 30% to the pipes flow rate.



3.8. Vacuum.

- ❑ The **suction pad** is the final element used to move pieces using vacuum. It is a simple, effective and economical device used for the movement of different types of materials and it is always an alternative to more sophisticated and expensive methods.
- ❑ The pad might lift, move and pick up elements with different weights, from a few grams up to 100 Kg.
- ❑ Its selection should be done according to the following variables:
 - The weight to be lifted.
 - The object position (horizontal or vertical)
 - The shape.
 - The material's compatibility.
 - The lineal speed or angular rotational speed and the braking options.



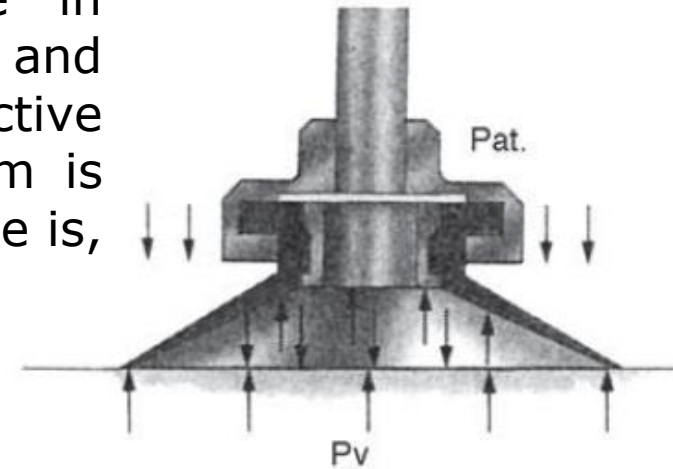


3.8. Vacuum.

- The **theoretical suction force** that provides a suction pad comes from the difference in pressure between the atmospheric (outside) and vacuum generated (inside), times the effective surface of the pad. The higher the vacuum is (lower pressure), the higher the exerted force is, according to:

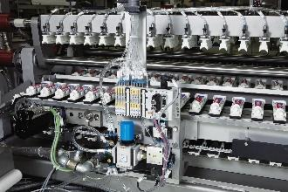
$$F_T = 1.033 p_v \left(\frac{\pi D^2}{400} \right) 9.8 \text{ (N)}$$

p_v in kg/cm².
D in mm.



Theoretical suction Force (N)

Ø Ventosa			2	4	6	10	13	16	20	25	32	40	50	63	80	100	125
A: Area de la ventosa (cm ²)			0.031	0.125	0.283	0.785	1,33	2.01	3.14	4.91	8.04	12.60	19.60	31,15	50,24	78,5	122,65
Vacío (mbar/mmHg) 1 mmHg=1,333 mbar	-866	-650	0.27	1.09	2.45	6.80	11,70	17.40	27.20	42.50	69.70	109.20	169.80	270,10	435,40	680,30	1.062,90
	-800	-600	0.25	0.98	2.26	6.30	10,80	16.10	25.10	39.30	64.30	100.80	156.70	249,20	401,90	627,90	981,20
	-733	-550	0.23	0.92	2.07	5.80	9,99	14.70	23.00	36.00	58.90	92.40	143.70	228,40	368,40	575,60	899,40
	-666	-500	0.21	0.84	1.89	5.20	9,90	13.40	20.90	32.70	53.60	84.00	130.60	207,70	334,90	523,30	817,60
	-600	-450	0.19	0,76	1.70	4.70	8,10	12.10	18.80	29.50	48.20	75.60	117.60	186,90	301,40	471,10	735,90
	-533	-400	0,17	0.67	1.51	4.20	7,20	10.70	16.70	26.20	42.90	67.20	104.50	166,10	267,90	418,60	654,10
	-467	-350	0.14	0.59	1.32	3.70	6,30	9.40	14.60	22.90	37.50	58.80	91.50	145,40	234,40	366,30	572,38
	-400	-300	0.12	0.50	1.13	3.14	5,40	8.00	12.60	19.60	32.10	50.40	78.40	124,60	200,90	313,90	490,60



3.8. Vacuum.

❑ Nevertheless, for the calculation of the real elevation force, the following parameters must be considered:

- Position (horizontal/vertical) of the weight to be lifted.
- Linear or rotational speed.
- Braking.

❑ And, combining the theoretical results and the experimental data, a security coefficient can be fixed (C) to obtain the **real force** to be able to lift an object, namely:

$$F_R = \frac{F_T}{C}$$

Suction pad

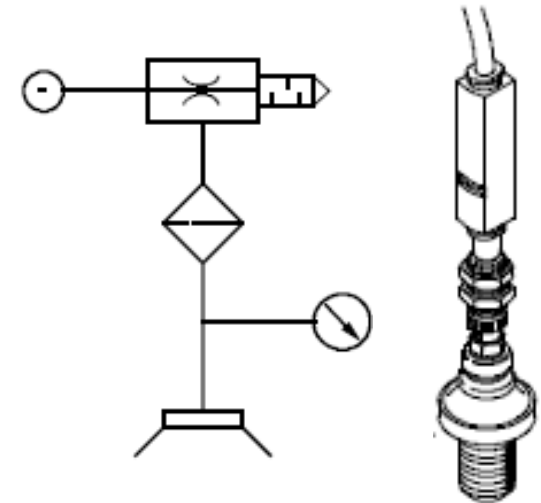


Horizontal weight

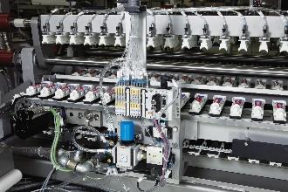


Vertical weight

Typical system with ejector and suction pad



Security coefficient, C	Horizontal movement	Vertical movement
Dynamic	≥ 4	≥ 8
Static	≥ 2	≥ 4



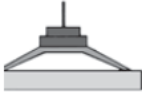
3.8. Vacuum.

- ❑ **Dimensions.** Calculation of the suction pad surface to hold a given load (N suction pads):

p_v in kg/cm².
W in kg.

$$D = 10 \sqrt{\frac{4C}{\pi N} \frac{W}{p_v}} \text{ (mm)}$$

- ❑ **Geometric shapes:**

Fixed surfaces		Plane surfaces
Surfaces with possible deformation		Plane ribbed surface
Curved surfaces		Concave surfaces
Inclined surfaces and height compensation		Bellows

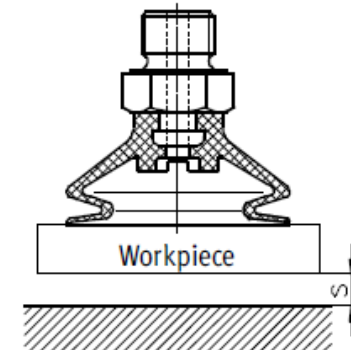
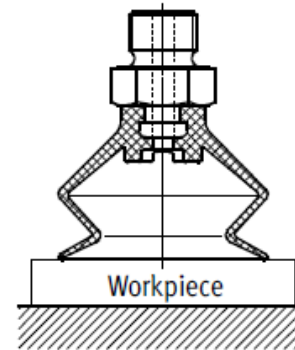
Forma

Forma de ventosa	Aplicaciones
Plana 	Usar cuando la superficie de la pieza es buena, plana y no tiene deformaciones.
Plana con nervios 	Usar cuando la pieza tiene tendencias a deformaciones.
Cóncava 	Utilizar cuando la superficie de la pieza es curvada.
Fuelles 	Utilizar cuando no hay espacio para montar telescópico y la pieza está inclinada.
Ventosa elipse 	Para una pieza de trabajo con una superficie pequeña de adsorción o para localizar con precisión una pieza larga.
Ventosa rótula 	Para una pieza de trabajo con superficie de adsorción no horizontal.
Telescópico de carrera larga 	Para pieza de trabajo con una altura no uniforme o con amortiguación necesaria.
Telescópico de gran tamaño 	Pieza pesada.
Ventosa conductiva 	Para la prevención de electricidad estática. Se utiliza material de goma con reducida resistencia como medida de prevención.



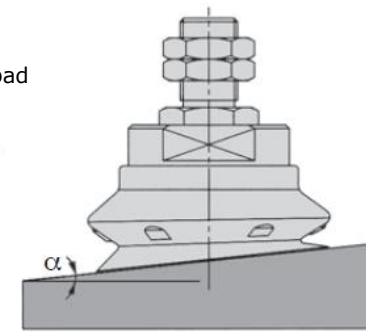
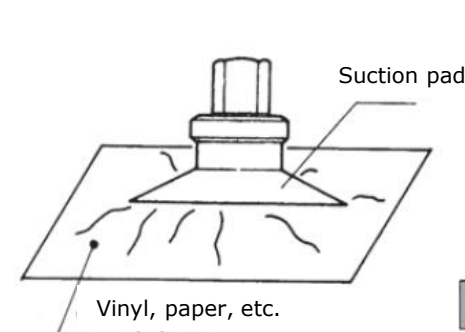
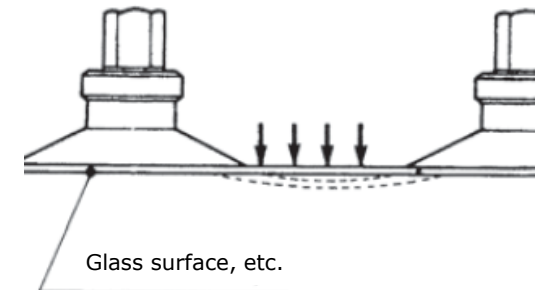
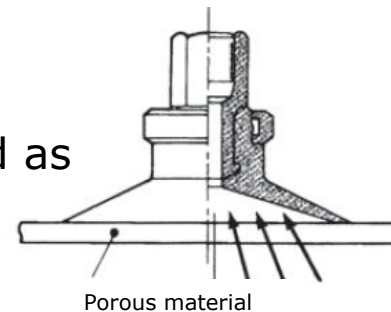
3.8. Vacuum.

❑ **Use and mounting:** It is recommended that the suction pad takes contact with the piece without sudden movements or high pressure, within the deformation margin of the pad material. **Always in a well centred setting.**



❑ **Complex objects:**

- Porous surfaces: diameter of the pad as small as possible.
- Plane surfaces: certain trend to separation: placement and size of the pad well defined, with possible inclination.
- Soft materials: pad as small as possible, using ribs and reducing the vacuum degree.
- Inclined elements: never for angles higher than 5° .





3.9. Bibliography and additional material.

❑ Books:

- **"Neumática", 2002.** SMC International Training. 2º Ed. Thomson-Paraninfo.
- Esposito, A., 2008; **"Fluid Power with Applications"**. Ed. Pearson
- **"Curso de Neumática Básica"**, 2007. CTAI Ingeniería.
- **"Fundamentos de Neumática"**, 2000, Festo Didactic GmbH&Co.

❑ On-line material:

- Introduction to the **pneumatics** and its **basic components**:
<http://industrial-automatica.blogspot.com/p/neumatica.html>
- **Vacuum systems.** Dimensions and analysis:
[http://www.smc.eu/portal/NEW_EBP/12\)Vacuum_Equipment/12.1\)Selection_Procedure/a\)Selection_Procedure/Vaccumequip_ES.pdf](http://www.smc.eu/portal/NEW_EBP/12)Vacuum_Equipment/12.1)Selection_Procedure/a)Selection_Procedure/Vaccumequip_ES.pdf)
http://www.festo.com/net/SupportPortal/Files/286804/Basic_Vacuum_Technology_Principles.pdf